

5G RAN

URLLC Feature Parameter Description

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1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes
None
- Editorial changes
Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 06 (2025-12-14).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added the impact relationship between "identification of UEs requiring connectivity with low latency and high reliability" and "QCI-specific uplink data compression". For details, see 4.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added the mutually exclusive relationship between FSD transmission mode and dynamic RB control. For details, see 6.1.3.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
Added the mutually exclusive relationship between identification of UEs requiring connectivity with low latency and high reliability and RF module fast deep dormancy. For details, see 4.3.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added the mutually exclusive relationship between identification of UEs requiring connectivity with low latency and high reliability and optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module deep dormancy. For details, see 4.3.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added the mutually exclusive relationship between identification of UEs requiring connectivity with low latency and high reliability and optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module super dormancy. For details, see 4.3.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added the impact relationships of "identification of UEs requiring connectivity with low latency and high reliability" with the following functions: uplink precise MCS optimization, probability-based uplink MU-MIMO MCS selection, and multi-beam reception enhancement. For details, see 4.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added the impact relationships of time-domain power saving BWP with identification of UEs requiring connectivity with low latency and high reliability. For details, see 4.3.2.3 Function Impacts .	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added "policies for controlling the number of UEs for preallocation". For details, see <i>Service-differentiated Experience Guarantee</i>. Added suggestions for setting PRESCH_UE_NUM_CTRL_POL_SW. For details, see: • 5.1.1 Principles • 5.1.4.1.1 Data Preparation • 5.1.4.1.2 Using MML Commands	Added the NRDUCellUlsch.QciPreschUeNumLimit and NRDUCellUlsch.CellPreschUeNumLimit parameters. Modified parameter: Added the PRESCH_UE_NUM_CTRL_POL_SW option to the NRDUCellUlsch.UlPreallocPolEnhSwitch parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added support for low latency and high reliability by UMPThb series boards. For details, see the "Hardware" sections in this document.	None	Low-frequency TDD	5900 series base stations (macro base stations) DBS5900 LampSite base stations

- Editorial changes
 - Revised the description of the impact relationship between identification of UEs requiring connectivity with low latency and high reliability and intelligent power control energy saving. For details, see [4.3.2.3 Function Impacts](#).
 - Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed. Examples of such changes include revising descriptions of cell requirements for certain functions.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

Trial Features

Trial features are features that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these features can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial

features shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial features. Trial features are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial features may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial features but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial features are free, Huawei is not liable for any trial feature malfunctions or any losses incurred by using the trial features. Huawei does not promise that problems with trial features will be resolved in the current version. Huawei reserves the rights to convert trial features into commercial features in later R/C versions. If trial features are converted into commercial features in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial features. If a customer fails to purchase such a license, the trial feature(s) will be invalidated automatically when the product is upgraded.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FOFD-05120 6	URLLC Introduction (Trial)	4 Low Latency and High Reliability Basics 5.2 Uplink Segmentation-Free Scheduling 6.1 Data Channel Reliability Optimization 6.2 Control Channel Reliability Optimization

3 Overview

Ultra-reliable low-latency communication (URLLC) is distinguished by its capability to provide connectivity with ultra-low latency and ultra-high reliability. It represents one of the three application scenarios of 5G and is applicable to a wide range of industries.

Latency, a crucial indicator for URLLC, can be quantified using either of the following metrics:

- End to end one-trip time (E2E OTT): refers to the time required for a signal to be transmitted from a UE to a server (located on the core network) or from a server to a UE. The calculation of this metric is challenging due to the lack of time synchronization between the server and the UE. Consequently, another metric, end to end round-trip time (E2E RTT), is usually calculated instead.
- E2E RTT: refers to the time required for a signal to be transmitted from a UE to a server plus the time required for a response to be transmitted back from the server to the UE. E2E RTT includes RAN RTT (which refers to the RTT between the UE and the gNodeB) and non-RAN RTT (which refers to the RTT between the gNodeB and the core network). In this document, only RAN RTT is discussed.

Two principal indicators are used to measure URLLC network performance: the maximum RTT and RTT variation. The following explanations are provided for these indicators:

- The maximum RTT is determined by sorting all RTT samples in ascending order within a test. The RTT corresponding to the 99.99% point in a cumulative distribution function (CDF) is then selected as the maximum RTT.
- RTT variation = Maximum RTT – Minimum RTT

The features and functions described in this document are only applicable to the business (ToB) sector and special wide area networks (WANs) and not to common WANs. An illustrative application scenario is a power grid network, which serves a small number of UEs requiring connectivity with high reliability. The features and functions are classified into three categories based on their underlying mechanisms, as described below:

- Some features and functions are oriented toward providing connectivity with low latency. This is achieved through mechanisms that reduce the latency introduced in the scheduling request loop, uplink and downlink

retransmissions, and other procedures. For details, see [3.1 Overview of Features and Functions Oriented Toward Delay Reduction](#).

- Some features and functions are oriented toward providing connectivity with high reliability. This is achieved through mechanisms that improve the reliability of data and control channels. For details, see [3.2 Overview of Features and Functions Oriented Toward Reliability Improvement](#).
- Some features and functions are oriented toward supporting connectivity with low latency and high reliability on an increased number of UEs. This is achieved through mechanisms that reduce resource consumption and improve resource utilization. For details, see [3.3 Overview of Features and Functions Oriented Toward Increasing the Number of Supported UEs Requiring Connectivity with Low Latency and High Reliability](#).

For details about the features and functions that are used in vehicle-to-everything (V2X) communications, see *URLLC for Vehicle Communications*.

NOTE

To facilitate differentiated processing for UEs requiring connectivity with low latency and high reliability, the base station must distinguish such UEs from common UEs. With this in mind, QCI or 5QI must differ between the two types of UEs. Moreover, certain parameters must be configured on the base station for it to identify UEs requiring connectivity with low latency and high reliability. In this document, the following descriptions are used interchangeably: UEs requiring connectivity with low latency and high reliability, UEs requiring connectivity with high reliability, and URLLC UEs. For details about identification of UEs requiring connectivity with low latency and high reliability, see [4.1.1 Identification of UEs Requiring Connectivity with Low Latency and High Reliability](#).

3.1 Overview of Features and Functions Oriented Toward Delay Reduction

Table 3-1 Features and functions oriented toward delay reduction

Feature/ Function Name	Description	Chapter/ Section	Dependency on Identification of UEs Requiring Connectivity with Low Latency and High Reliability
Service-differentiated low latency	This function enables the differentiation of scheduling parameter configuration between QCI. This reduces the processing time required for scheduling, the block error rate (BLER), and the number of retransmissions, thereby decreasing the delay over the air interface.	5.1 Service-Differentiated Low Latency	No
PDCCP out-of-order delivery	This function allows out-of-order packets to be transmitted to the upper layer for processing in the event of packet disorder, provided that the receiver supports Packet Data Convergence Protocol (PDCCP) out-of-order delivery. This process does not involve PDCCP reordering, which reduces delay.	PDCCP Out-of-Order Delivery	Yes

Feature/ Function Name	Description	Chapter/ Section	Dependency on Identification of UEs Requiring Connectivity with Low Latency and High Reliability
HARQ retransmission optimization	This function applies an upper limit to the number of HARQ retransmissions. If the number of HARQ retransmissions for a data packet exceeds the upper limit, the data packet will be discarded. This mechanism prevents extra resource consumption and prevents such data packets from impeding the transmission and reception of other data packets, thereby shortening the delay caused by waiting.	HARQ Retransmission Optimization	Yes
Uplink segmentation-free scheduling	This function increases the uplink MCS index by a fixed value of 3 for UEs with delay-sensitive factors configured. This mechanism reduces the number of uplink fragments or prevents uplink segmentation, thereby shortening the delay caused by fragmentation.	5.2 Uplink Segmentation-Free Scheduling	Yes

3.2 Overview of Features and Functions Oriented Toward Reliability Improvement

Table 3-2 Features and functions oriented toward reliability improvement

Feature/ Function Name	Description	Chapter/ Section	Dependency on Identification of UEs Requiring Connectivity with Low Latency and High Reliability
Data channel reliability optimizat ion	• Resource reservation: This function guarantees the availability of data channel resources for services that require connectivity with high reliability. • Outer loop link adaptation (OLLA) algorithm optimization: The MCS index range can be optimized to reduce the BLER in the event of signal fluctuation. • CQI table 3: UEs can report measurement results according to CQI table 3, thereby reducing the BLER of services requiring connectivity with high reliability. • MCS index table 3: The table incorporates low code rates and can be used to reduce the BLER of services requiring connectivity with high reliability.	6.1 Data Channel Reliability Optimization	Yes

Feature/ Function Name	Description	Chapter/ Section	Dependency on Identification of UEs Requiring Connectivity with Low Latency and High Reliability
Control channel reliability optimization	- Target PDCCH IBLER optimization: The target initial block error rate (IBLER) can be reduced to increase the PDCCH signal to interference plus noise ratio (SINR), thereby improving PDCCH reliability. - PDCCH CCE optimization: This function incorporates the CCE resource increase, uplink-to-downlink CCE ratio adaptation, and SR-based smart scheduling sub-functions. It is recommended that these sub-functions be enabled when there are a large number of online UEs. - PDCCH aggregation level adjustment mechanism optimization: The minimum PDCCH aggregation level can be increased to a proper level. This increases blind detections with higher PDCCH aggregation levels while restricting the number of blind detections to the upper limit. Consequently, the probability of missed PDCCH detections is reduced.	6.2 Control Channel Reliability Optimization	Yes

3.3 Overview of Features and Functions Oriented Toward Increasing the Number of Supported UEs Requiring Connectivity with Low Latency and High Reliability

Table 3-3 Features and functions oriented toward increasing the number of supported UEs requiring connectivity with low latency and high reliability

Feature/ Function Name	Description	Chapter/ Section	Dependency on Identification of UEs Requiring Connectivity with Low Latency and High Reliability
PLC UE number specificati on improvement	Self-learning-based preallocation is configured to allow the gNodeB to adaptively learn the time at which a UE transmits a service packet, calculate an interval between packet transmissions, and perform intermittent preallocation according to the interval. This reduces uplink resource consumption, which in turn increases the number of online programmable logic controller (PLC) UEs supported on a network.	7.1 PLC UE Number Specification Improvement	Yes
Scheduling based on delay- sensitive requirements	This function enables the differentiation of delay guarantee between UEs requiring connectivity with high reliability, thereby alleviating uplink resource insufficiency.	7.2 Scheduling Based on Delay- Sensitive Requirements	Yes

4 Low Latency and High Reliability Basics

When a UE accesses a network, the gNodeB can identify whether the UE requires connectivity with low latency and high reliability based on the UE's QCI. Once a UE is identified as requiring connectivity with low latency and high reliability, features and functions aimed at reducing latency and improving reliability can be implemented to ensure low latency and high reliability for the UE.

4.1 Principles

4.1.1 Identification of UEs Requiring Connectivity with Low Latency and High Reliability

The QCIs or 5QIs planned for common UEs are different from those for UEs requiring connectivity with low latency and high reliability. This differentiation facilitates the identification of a UE requiring connectivity with low latency and high reliability. Further details are provided below.

1. The core network assigns QCIs or 5QIs to UEs.

The QCIs or 5QIs assigned by the core network differ between common UEs and UEs requiring connectivity with low latency and high reliability. During RAB setup, the core network notifies the corresponding gNodeB of QCIs or 5QIs that are allocated to each type of UE. In SA networking, after receiving 5QIs assigned for a type of UE, the gNodeB maps the 5QIs to the corresponding QCIs.

2. The gNodeB identifies whether a UE requires connectivity with low latency and high reliability. This identification requires the **HIGH_RELIABILITY_BASIC_SW** option of the **NRDUCellAlgoSwitch.HighReliabilitySwitch** parameter to be selected. For a UE with a specific QCI to be identified as requiring connectivity with low latency and high reliability, all the following QCI-specific configuration is required:
 - The **gNBDUMacParamGroup.SchedulingType** parameter is set to **HIGH_RELIABILITY_SCH**. This enables high-reliability-oriented scheduling for the QCI-specific bearer.

- The **gNBQciBearer.ServiceType** parameter is set to **URLLC_HIGH_RELIABILITY_SERVICE**. This enables the QCI-specific bearer to carry services requiring connectivity with high reliability.
- 3. If a QCI-specific bearer has been set up for a UE to carry a service requiring connectivity with high reliability, the gNodeB identifies the UE as requiring connectivity with low latency and high reliability. A UE that has been identified as requiring connectivity with low latency and high reliability has to re-access the network if the **gNBQciBearer.ServiceType** parameter value for a QCI that belongs to the UE is changed on the base station but the corresponding bearer remains unchanged on the UE. Doing so will make the said change take effect; failure to do so will result in the UE type remaining unchanged.

Inter-frequency measurement in a measurement gap results in a deterioration in the performance of services requiring connectivity with high reliability. For this reason, it is recommended that the **NRCellInterFHoMeaGrp.CovInterFreqA2RsrpThld** and **NRCellInterFHoMeaGrp.FreqPriInterFA1RsrpThld** parameters be set to **-110** and **-31**, respectively, for the QCIs of the bearers set up for low latency and high reliability.

NOTE

UEs perform inter-frequency measurements for a variety of handovers, including coverage-based inter-frequency handovers, frequency-priority-based inter-frequency handovers, and service-based inter-frequency handovers.

In cells where services requiring connectivity with low latency and high reliability are running, load balancing causes a decline in the performance of these services. For this reason, it is recommended that the **NRCellMlb.QciSwBasedUeSelPolicyPri** parameter be set to **255** and the **MLB_FILTER_SW** option of the **NRCellQciBearer.MlbAlgoSwitch** parameter be selected.

4.1.2 Admission Control of Low Latency and High Reliability

The 'admission control of low latency and high reliability' function enables admission control on UEs requiring connectivity with low latency and high reliability. This function ensures the optimal utilization of limited resources by such UEs. **A UE requiring connectivity with low latency and high reliability can be admitted if the UE has only one bearer.**

If multiple bearers are configured for a UE requiring connectivity with low latency and high reliability, the following rules apply:

- If none of these bearers have a QCI of 1 or 5, the UE's bearer setup request and cell access request will be rejected.
- If some of these bearers have a QCI of 1 or 5, the UE's bearer setup request and cell access request will be accepted, but low latency and high reliability will not be implemented.

After specific parameters are modified, RRC connection reconfiguration must be proactively initiated (by running the **STR NRCELLSERVICEPARAMEFFECT** command) to make the modification take effect on online UEs in real time. Once the RRC connection reconfiguration is completed, the base station checks whether

the online UEs can be identified as requiring connectivity with low latency and high reliability. If so, the online UEs can be admitted provided that they have only one bearer. If not, the online UEs are considered as common UEs in admission control. The aforementioned parameters are as follows:

- **HIGH_RELIABILITY_BASIC_SW** option of the **NRDUCellAlgoSwitch.HighReliabilitySwitch** parameter
- **gNBDUMacParamGroup.SchedulingType**

4.2 Network Analysis

4.2.1 Benefits

Identification of UEs requiring connectivity with low latency and high reliability serves as the basis for other features and functions related to low latency and high reliability to take effect.

Admission control of low latency and high reliability restricts the number of UEs requiring connectivity with low latency and high reliability that are permitted to access a cell, thereby ensuring the optimal utilization of limited resources by UEs requiring connectivity with low latency and high reliability.

Conditions for Optimal Gains

Optimal gains can be delivered by the features and functions that rely on identification of UEs requiring connectivity with low latency and high reliability when all of the following conditions are met:

- Cell coverage: RSRP > -80 dBm; interference over thermal (IoT) < 10 dB
- Network load: light network load; cell uplink RB usage < 40%; cell downlink RB usage < 40%; maximum RB usage < 80%
- Mobility: stationary UEs or slow-moving UEs not involved in handovers
- Traffic model: small-packet services with discontinuous data transmission

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

If a UE is configured with more than one bearer, and the bearers of the UE include high-reliability-oriented bearers but exclude bearers of QCI 1 or QCI 5, the base station rejects the UE's bearer setup request or request for accessing a high-reliability-oriented cell. For handovers from other types of cells to high-reliability-oriented cells for such UEs, the handover success rate decreases.

NOTE

If the **HIGH_RELIABILITY_BASIC_SW** option of the **NRDUCellAlgoSwitch.HighReliabilitySwitch** parameter is selected for a cell, the cell is a high-reliability cell.

If the **HIGH_RELIABILITY_BASIC_SW** option of the **NRDUCellAlgoSwitch.HighReliabilitySwitch** parameter is deselected for a cell, the cell is not a high-reliability cell.

4.3 Requirements

4.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-051206	URLLC Introduction	NR0SURLLCP00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Identification of UEs requiring connectivity with low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter gNBDUMacParamGroup.SchedulingType set to HIGH_RELIABILITY_SCH gNBQciBearer.ServiceType set to URLLC_HIGH_RELIABILITY_SERVICE	<i>URLLC</i>	'Admission control of low latency and high reliability' depends on 'identification of UEs requiring connectivity with low latency and high reliability'.

4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag set to HIGH_SPEED	<i>High Speed Mobility</i>	The HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliability-Switch parameter cannot be selected in high-speed mobility scenarios.
Low-frequency TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	The HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliability-Switch parameter and the RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter cannot be both selected.
Low-frequency TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	The HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliability-Switch parameter and the TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter cannot be both selected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intelligent carrier shutdown	INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	The HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliability-Switch parameter and any option that controls intelligent carrier shutdown cannot be both selected.
Low-frequency TDD	Common-channel dynamic shutdown in TTI-level channel shutdown	NRDUCellPowerSaving.CommonChnDynShutdownSwitch and TTI_RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When the HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliability-Switch parameter is selected, common-channel dynamic shutdown in TTI-level channel shutdown cannot be enabled.
Low-frequency TDD	LTE and NR intelligent carrier shutdown	LNR_SMART_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	The HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliability-Switch parameter and the LNR_SMART_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter cannot be both selected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inter-FR CA	INTER_FR_FR1TOFR2_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	The INTER_FR_FR1TOFR2_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter and the HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliability-Switch parameter cannot be both selected.
High-frequency TDD	Full-buffer UE-specific CSI-RS for BM	USER_BM_CSIRS_SW option of the NRDUCellCsirs.CsiSwitch parameter	<i>Channel Management</i>	When the HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliability-Switch parameter is selected, the USER_BM_CSIRS_SW option of the NRDUCellCsirs.CsiSwitch parameter cannot be selected.
Low-frequency TDD	PUSCH Sch in S Slot Sw	PUSCH_SCH_IN_S_SLOT_SW option of the NRDUCellPusch.UlPuschAlgoSwitch parameter	None	The HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliability-Switch parameter and the PUSCH_SCH_IN_S_SLOT_SW option cannot be both selected.
Low-frequency TDD	Uplink symbol power saving	UL_SYMBOL_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	The two functions—(1) uplink symbol power saving and (2) low latency and high reliability—are mutually exclusive.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fast carrier shutdown	- Capacity-layer cells: - Intra-base-station multi-carrier scenario: : INTRA_GNB_MULTICARRIER_SWITCH option of the NRDUCellAlgoSwitch. <i>PowerSavingSwitch</i> parameter - Inter-base-station multi-carrier scenario: : INTER_GNB_MULTICARRIER_SWITCH option of the NRDUCellAlgoSwitch. <i>PowerSavingSwitch</i> parameter FAST_CARRIER	<i>Energy Conservation and Emission Reduction</i>	The low latency and high reliability function does not work with fast carrier shutdown.

RAT	Function Name	Function Switch	Reference	Description
		_SHUT_DOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter Basic-layer cells: FAST_CARRIER_SHUT_SERV_ASSUR_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter		
Low-frequency TDD	Common-channel dynamic shutdown in dynamic RF channel shutdown	NRDUCellPowerSaving.CommonChnDynamicShutdownSwitch and RF_CHN_DYN_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When the HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter is selected, common-channel dynamic shutdown in dynamic RF channel shutdown cannot be enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Single DCI	DL_SINGLE_DCI_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgorithmSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	The HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliability-Switch parameter and the DL_SINGLE_DCI_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgorithmSwitch parameter cannot be both selected.
Low-frequency TDD	QoS Class Identifier	NRDUCellOpQci.Qci	<i>QCI-based Resource Control</i>	A QCI cannot be bound to the NRDUCellOpQci MO if the QCI meets either of the following conditions: - The gNBDUMacParamGroup.SchedulingType parameter associated with the QCI is set to HIGH_RELIABILITY_SCH. - The gNBQciBearer.ServiceType parameter associated with the QCI is set to URLLC_HIGH_RELIABILITY_SERVICE.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF module deep dormancy	RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB) set to ON or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	In low latency and high reliability scenarios, the following functions cannot be enabled: NR load-based RF module deep dormancy, LNR load-based RF module deep dormancy, and optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module deep dormancy.
Low-frequency TDD	RF module super dormancy	RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB) set to ON_SUPER or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	In low latency and high reliability scenarios, the following functions cannot be enabled: NR load-based RF module super dormancy, LNR load-based RF module super dormancy, and optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module super dormancy.
Low-frequency TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATSW (5G gNodeB, LTE eNodeB) parameter	<i>Multi-RAT Coordinated Energy Saving</i>	NR and LNR load-based RF module fast deep dormancy are not supported in low latency and high reliability scenarios.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Master TRP of a Light DM MIMO Cell	NRDUCellTrp.TrpType set to MASTER_DM_MIMO_LIGHT	None	When the NRDUCellTrp.TrpType parameter is set to MASTER_DM_MIMO_LIGHT for a cell, the gNBQciBearer.ServiceType parameter cannot be set to URLLC_HIGH_RELIABILITY_SERVICE for the cell.
Low-frequency TDD	Periodic connected mode load balancing	QCI_UE_NUM_BASED_PRDC_BLB_SW option of the gNBDURedCapBlbGrp.RedCapBlbAlgoSwitch parameter	<i>RedCap</i>	The QCI_UE_NUM_BASED_PRDC_BLB_SW option of the gNBDURedCapBlbGrp.RedCapBlbAlgoSwitch parameter cannot be selected for a parameter group bound to a QCI indicating high-reliability services (with the gNBQciBearer.ServiceType parameter set to URLLC_HIGH_RELIABILITY_SERVICE).

4.3.2.3 Function Impacts

- Functions that are more likely to take effect than low latency and high reliability

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Super uplink	SUPER_UL_TDM_SCH_SW or FLEX_SUPER_UPLINK_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter	<i>Super Uplink</i>	The low latency and high reliability function does not take effect for UEs enabled with super uplink.
Low-frequency TDD	UL and DL Decoupling	NRDUCellAlgoSwitch.ULDLDecouplingSwitch	<i>UL and DL Decoupling</i>	The low latency and high reliability function does not take effect for UEs using UL and DL Decoupling.
Low-frequency TDD	VoNR	VONR_SW option of the NRCelAlgoSwitch.VonrSwitch parameter	<i>VoNR</i>	When the VoNR function takes effect for a UE, the low latency and high reliability function is terminated.

- Functions that are less likely to take effect than low latency and high reliability

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UL 256QAM	NRDUCellAlgoSwitch.UL256QamSwitch set to UL_256QAM_FIXED or UL_256QAM_ADAPT	<i>Modulation Schemes</i>	UL 256QAM does not take effect for UEs requiring connectivity with low latency and high reliability.
Low-frequency TDD	DL 256QAM	NRDUCellAlgoSwitch.DL256QamSwitch	<i>Modulation Schemes</i>	DL 256QAM does not take effect for UEs requiring connectivity with low latency and high reliability.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	MU-MIMO pairing	NRDUCellAIgoSwitch.MuMimoSwitch	<i>MIMO (TDD)</i>	UEs requiring connectivity with low latency and high reliability cannot be paired for MU-MIMO.
Low-frequency TDD	Power saving BWP	BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	When the low latency and high reliability function and power saving BWP are both enabled, UEs requiring connectivity with low latency and high reliability do not switch to BWP2.
Low-frequency TDD	Uplink preallocation for sparse-packet services	SPARSE_DIFF_PROC_SW option of the gNodeBParam.ServiceDiffProcSwitch parameter and SPARSE_DIFF_SCH_SW option of the NRDUCellAIgoSwitch.ServiceDiffSwitch parameter	<i>Uplink Scheduling</i>	Uplink preallocation for sparse-packet services does not take effect for UEs requiring connectivity with low latency and high reliability.
Low-frequency TDD	Uplink basic preallocation	UL_BASIC_PREALLOCATION_SWITCH option of the NRDUCellPusch.UlPreallocationSwitch parameter	<i>Uplink Scheduling</i>	When the low latency and high reliability function and uplink basic preallocation are both enabled for services with a specific QCI, uplink basic preallocation does not take effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink smart preallocation	UL_SMART_PREALLOCATION_SWITCH option of the NRDUCellPusch.UlPreallocationSwitch parameter	<i>Uplink Scheduling</i>	When the low latency and high reliability function and uplink smart preallocation are both enabled for services with a specific QCI, uplink smart preallocation does not take effect.
Low-frequency TDD	Secondary harmonic interference avoidance	NRDUCellNsAdcConfig.HarmonicIntrfAvoidRange and HARMONIC_INTRF_AVOID_SW option of the NRDUCellAlgoSwitch.NsAdcAlgoSwitch parameter	<i>EPC-based NSA Basics</i>	Secondary harmonic interference avoidance does not take effect if it is enabled together with the low latency and high reliability function.
Low-frequency TDD	UE-number-based connected mode MLB	INTER_FREQ_CONNECTED_MLB_SW option of the NRCCellAlgoSwitch.MlbAlgoSwitch parameter, and NRCCellMlb.MlbTriggerMode set to CELL_USER_NUM	<i>Mobility Load Balancing in Connected Mode</i>	The gNodeB does not select UEs requiring connectivity with low latency and high reliability for handovers.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Spectral-efficiency-based connected mode MLB	INTER_FREQ_CONNECTE D_MLB_SW option of the NRCCellAlgoS witch.MlbAl goSwitch parameter, and NRCCellMlb. MlbTrigger Mode set to SPEC_EFF_B ASED_USER_ NUM	<i>Mobility Load Balancin g in Connecte d Mode</i>	The gNodeB does not select UEs requiring connectivity with low latency and high reliability for handovers.
Low-frequency TDD	5QI-specific UE-number-based connected mode MLB	INTER_FREQ_CONNECTE D_MLB_SW option of the NRCCellAlgoS witch.MlbAl goSwitch parameter, and NRCCellMlb. MlbTrigger Mode set to SPECIFIED_5 QI_USER_NU M	<i>Mobility Load Balancin g in Connecte d Mode</i>	The gNodeB does not select UEs requiring connectivity with low latency and high reliability for handovers.
Low-frequency TDD	PDCCH skipping	PDCCH_SKIP _SW option of the NRDUCellUe PwrSaving.S chPwrSavin gSw parameter	<i>UE Power Saving</i>	When the low latency and high reliability function and PDCCH skipping are both enabled, PDCCH skipping does not take effect for the UEs requiring connectivity with low latency and high reliability.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Power Saving Switch	NRDUCellAlgoSwitch.PowerSavingSwitch	<i>Energy Conservation and Emission Reduction</i> <i>Multi-RAT Coordinated Energy Saving</i>	If the low latency and high reliability function is enabled, it is recommended to disable functions related to base station power saving (by deselecting related options of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter). - The power saving functions that are separately controlled by the TTI_RF_SHUTDOWN_SW, DATA_CONVERGENCE_IN_SLOTS_SW, and FLEX_FREQ_SCH_ENERGY_SAVING_SW options do not take effect for UEs requiring connectivity with low latency and high reliability. - When other options are selected, there may be less decrease in delay.
Low-frequency TDD	Downlink intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAlgoSwitch.CarrierAggregation parameter	<i>Carrier Aggregation</i>	Downlink intra-band CA does not take effect for UEs requiring connectivity with low latency and high reliability.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellInfoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	Downlink intra-FR inter-band CA does not take effect for UEs requiring connectivity with low latency and high reliability.
Low-frequency TDD	Uplink data compression	UDC_SW option of the NRDUCellChnCovAlgo.UlCoverageAlgoSwitch parameter, UPLINK_SIG_COMPRESS_SW option of the NRDUCellChnCovAlgo.UlCoverageAlgoSwitch parameter, or QCI_UDC_SW option of the gNBQciBearer.PdcpAlgoSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	The low latency and high reliability function and uplink data compression cannot take effect simultaneously for the same bearer of a UE. The low latency and high reliability function preferentially takes effect for the bearer if it is enabled together with uplink data compression.
Low-frequency TDD	Probability-based uplink MCS selection	UL_PBMCSSEL_SW option of the NRDUCellUplinkAmc.UlAmcPerformanceSw parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	Probability-based uplink MCS selection does not take effect for UEs requiring connectivity with low latency and high reliability.
Low-frequency TDD	Dynamic RF channel shutdown	RF_CHN_DYNAMIC_SHUTDOWN_SW option of the NRDUCellInfoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Dynamic RF channel shutdown does not take effect for UEs requiring connectivity with low latency and high reliability.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DL 1024QAM	DL_1024QAM_SW option of the NRDUCellDLAmc.DLModulationSchemeSw parameter	<i>Modulation Schemes</i>	DL 1024QAM does not take effect for UEs requiring connectivity with low latency and high reliability.
Low-frequency TDD	Stable Video Surveillance Experience	gNBDUMacParamGroup.SchedulingType set to STABLE_UL_SERVICE_EX P_SCH	<i>Stable Video Surveillance Experience</i>	Stable Video Surveillance Experience does not take effect for UEs requiring connectivity with low latency and high reliability.
Low-frequency TDD	SSSG switching	UE_PWR_SAVING_ENHANCE_SW and SSSG_SWITCHING_SW options of the NRDUCellUePwrSaving.UePowerSavingSwitch parameter	<i>UE Power Saving</i>	UEs requiring connectivity with low latency and high reliability do not switch to a sparse search space set group (SSSG).
Low-frequency TDD	Time-domain power saving BWP	TDM_BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	When "low latency and high reliability" and time-domain power saving BWP are both enabled, the UEs using the "low latency and high reliability" function will not be switched to BWP2.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RedCap SSSG switching	REDCAP_SSSG_SWITCHING_SW or REDCAP_SSSG_SWITCH_EXT_SW option of the NRDUCellUePwrSaving.UePowerSavingSwitch parameter	<i>RedCap</i>	UEs requiring connectivity with low latency and high reliability do not switch to an SSSG.
Low-frequency TDD	Intelligent power control energy saving	SMART_POWER_CTRL_ENERGY_SAVE_SWITCH option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Intelligent power control energy saving does not take effect for UEs requiring connectivity with low latency and high reliability.
Low-frequency TDD	Uplink precise MCS optimization	UL_PRECISE_MCS_OPT_SWITCH option of the NRDUCellAIgo.AiAmcAlgoSwitch parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	Uplink precise MCS optimization does not take effect for UEs requiring connectivity with low latency and high reliability.
Low-frequency TDD	Probability-based uplink MU-MIMO MCS selection	UL_MU_PBM_CSEL_SW option of the NRDUCellUeAmc.UeAmcPerformanceSwitch parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	Probability-based uplink MU-MIMO MCS selection does not take effect for UEs requiring connectivity with low latency and high reliability.
Low-frequency TDD	Multi-beam reception enhancement	MULTI_BEAM_RX_ENH_SW option of the NRDUCellPusch.PuschAIgoExtSwitch parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	Multi-beam reception enhancement does not take effect for UEs requiring connectivity with low latency and high reliability.

- Other function impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DRX	BASIC_DRX_SW option of the NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch parameter	<i>DRX</i>	DRX is not recommended for bearers with the low latency and high reliability function enabled.
Low-frequency TDD	Service Type	gNBQciBearer.ServiceType set to URLLC_HIGH_RELIABILITY_SERVICE	None	On the base station, the gNBQciBearer.ServiceType parameter must be set to URLLC_HIGH_RELIABILITY_SERVICE for the QCI of a high-reliability-oriented bearer. Otherwise, low latency and high reliability will not take effect for the QCI-specific bearer.
Low-frequency TDD	Inter-frequency measurement	None	<i>SA Mobility Management in Connected Mode or NSA Mobility Management</i>	The performance of services requiring connectivity with high reliability deteriorates during UE inter-frequency measurements due to the start of gap-assisted measurements. UEs perform inter-frequency measurements for a variety of handovers, including coverage-based inter-frequency handovers, frequency-priority-based inter-frequency handovers, and service-based inter-frequency handovers.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Special Service PVI Mode	NRDUCellRac.SpecServPviMode set to PVI_FOLLOW_ARP	<i>Admission Control</i>	Resources allocated to a UE requiring connectivity with low latency and high reliability may be preempted by another UE that has a bearer of a higher ARP priority when these conditions are met: (1) The NRDUCellRac.SpecServPviMode parameter is set to PVI_FOLLOW_ARP , and (2) the UE requiring connectivity with low latency and high reliability has a bearer of a lower ARP priority and is marked as preemptable. To prevent this, it is recommended to configure the core network in such a way that UEs requiring connectivity with low latency and high reliability are given higher ARP priorities compared to other UEs and are marked as unpreemptable.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDCCH uplink-to-downlink CCE ratio adaptation	UL_DL_CCE_RATIO_ADAPT_SW option of the NrDuCellPdcch.PdcchAlgorithmSwitch parameter	<i>Channel Management</i>	When the 2_3_DSUUU slot configuration is used, it is recommended that PDCCH uplink-to-downlink CCE ratio adaptation be disabled and NRDUCellPdcch.UIMaxCcePct be set to 50 in scenarios with the low latency and high reliability function enabled.
Low-frequency TDD	CCE resource optimization	CCE_RESOURCE_OPT_SW option of the NRDUCellPdcch.PdcchAlgorithmEnhSwitch parameter	<i>MIMO (TDD)</i>	Enabling CCE resource optimization may reduce the reliability of the low latency and high reliability function.
Low-frequency TDD	Low power consumption mode	LOW_PWR_MODE_SW option of the NRDUCellLampSite.PowerSavingSwitch parameter	<i>Green Site</i>	After low power consumption mode is enabled for a cell, power saving features are enabled as required, affecting the performance of UEs requiring connectivity with high reliability.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

- Main control boards: Supported models are limited to the UMPTe/UMPTg/UMPTHa/UMPTHb series.
- Baseband processing units: Supported models are limited to the UBBPg and later series.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Cells

- Low-frequency NR TDD: The required cell bandwidth is 60 MHz, 80 MHz, or 100 MHz.

4.3.5 Others

None

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Table 4-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-1 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
High Reliability Switch	NRDUCellAlgoSwitch.HighReliabilitySwitch	HIGH_RELIABILITY_BASIC_SW	To enable low latency and high reliability, select this option.
Scheduling Type	gNBDUMacParamGroup.SchedulingType	None	To enable high-reliability-oriented scheduling, set this parameter to HIGH_RELIABILITY_SCH .

Parameter Name	Parameter ID	Option	Setting Notes
Service Type	gNBQciBearer.ServiceType	None	To enable high-reliability-oriented scheduling, set this parameter to URLLC_HIGH_RELIABILITY_SERVICE .

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

NOTICE

Before enabling low latency and high reliability, configure a high-reliability-oriented bearer for a QCI that is used for a service requiring connectivity with high reliability on the base station. Otherwise, low latency and high reliability will not take effect for the QCI-specific bearer.

QCI 9, QCI 1, and QCI 5 are not recommended for bearers that are used for low latency and high reliability.

- QCI 9 is used for the default bearer. If a QCI is configured in the **gNBQciBearer** MO but not configured in the **NRDUCellQciBearer** MO, the parameter settings related to QCI 9 are used by default.
- Bearers of QCI 1 and those of QCI 5 are dedicated for voice services.

Activation Command Examples

```
//Turning on the switch for basic high-reliability functions
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, HighReliabilitySwitch=HIGH_RELIABILITY_BASIC_SW-1;

//Adding a high-reliability-oriented bearer (Configuration with a single bearer of 5QI 82 requires support
from the core network.)
ADD GNB5QICONFIG: Nr5qi=82, Qci=82;
ADD GNBQCIBEARER: Qci=82;
//Setting gNBQciBearer.ServiceType to URLLC_HIGH_RELIABILITY_SERVICE for a QCI of a high-reliability-
oriented bearer on the base station (Otherwise, low latency and high reliability will not take effect for the
QCI-specific bearer.)
MOD GNBQCIBEARER: Qci=82, ServiceType=URLLC_HIGH_RELIABILITY_SERVICE;

//Configuring a service requiring connectivity with high reliability, with a high-reliability-oriented bearer
configured and associated with a QCI
ADD NRCELLQCIBEARER: NrCellId=1, Qci=82;
```

```
ADD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=82;  
MOD GNBDUMACPARAMGROUP: MacParamGroupId=1, SchedulingType=HIGH_RELIABILITY_SCH;  
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=82, MacParamGroupId=1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the switch for basic high-reliability functions  
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, HighReliabilitySwitch=HIGH_RELIABILITY_BASIC_SW-0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

In activation verification for identification of UEs requiring connectivity with low latency and high reliability, performance counters can be checked as follows:

- If the **HIGH_RELIABILITY_BASIC_SW** option of the **NRDUCellAlgoSwitch.HighReliabilitySwitch** parameter is selected and the value of the **N.QosFlow.Est.Succ.Cell5QI** counter (measuring the number of successful QoS flow setups for the QCI specific to the corresponding high-reliability-oriented bearer) is greater than 0, identification of UEs requiring connectivity with low latency and high reliability has taken effect. To obtain the average number of active UEs with the QCI specific to the corresponding high-reliability-oriented bearer, check the value of the **N.User.RRCCConn.Active.DuCell5QI.Avg** counter.
- The number of QoS flow setup failures for a QCI specific to the corresponding high-reliability-oriented bearer is calculated using the following formula: **N.QosFlow.Est.Att.Cell5QI – N.QosFlow.Est.Succ.Cell5QI**. To query the cause of the failures, perform the following steps:

Step 1 On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**.

Step 2 In the left navigation pane of the displayed **Signaling Trace Management** tab page, choose **NG Interface Trace**.

Step 3 On the displayed **NG Interface Trace** page, check the PDU SESSION RESOURCE SETUP RESPONSE message. Radio resources are unavailable in either of the conditions: (1) **PDU Session Resource Setup Unsuccessful Transfer** indicates a cause value of **Radio resources not available**; (2) **QoS Flow Item (PDU Session Resource Setup Response Transfer > QoS Flow Failed to Setup List)** indicates a cause value of **Radio resources not available** for the QoS flows corresponding to some services requiring connectivity with high reliability.

----End

4.4.3 Network Monitoring

Observing Delay by Segment

1. On the MAE-Evaluation home page, enter **Coverage Report** in the search box. In the **Search Results** area, click **Coverage Report**.
2. In the left pane of the **Coverage Report** page, set parameters based on actual conditions.
3. View the statistical result of the coverage report and monitor the uplink and downlink delay at the PDCP, RLC, and MAC layers.

Observing the BLER and MCS Index Statistics

Start a BLER monitoring task and an MCS count monitoring task on the MAE-Access.

To start BLER monitoring, perform the following steps:

1. On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**.
2. In the left navigation pane of the displayed **Signaling Trace Management** tab page, choose **User Performance Monitoring > BLER Monitoring** to observe the BLER.

To start MCS count monitoring, perform the following steps:

1. On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**.
2. In the left navigation pane of the displayed **Signaling Trace Management** tab page, choose **User Performance Monitoring > MCS Count Monitoring** to observe the average MCS index.

5 Delay Reduction

This section describes the functions that are oriented toward delay reduction.

5.1 Service-Differentiated Low Latency

5.1.1 Principles

Service-differentiated low latency can be enabled to reduce the air interface latency for UEs requiring connectivity with high reliability. This function is controlled by the **SERV_DIFF_LOW_LATENCY_SW** option of the **NRDUCellAlgoSwitch.LowLatencySwitch** parameter. For details about this function, see *Service-differentiated Experience Guarantee*. This function enables specific scheduling parameter settings for UEs requiring connectivity with high reliability to differ from other types of UEs. This reduces the processing time required for scheduling, the block error rate (BLER), and the number of retransmissions, thereby decreasing the air interface latency.

QCI-based Uplink Preallocation

With QCI-based uplink preallocation, a different preallocation parameter group can be configured for the bearer intended for URLLC, compared to other types of bearers. This reduces the processing time required for scheduling.

- QCI-based uplink basic preallocation
In QCI-based uplink basic preallocation, the gNodeB schedules a UE at regular intervals as long as resources are available, even if the UE does not have any service request. For this function to work in URLLC scenarios, the following parameters can be configured as required.

Item	Parameter	Setting Notes
Uplink Preallocation Interval	gNBDUSchParamGroup. <i>UlPreallocationInterval</i>	TDD: - Set this parameter to 5 for periodic services that require a 20 ms E2E delay. - Set this parameter to 40 for periodic services that require a 50 ms E2E delay.
Uplink Preallocation Data Volume	gNBDUSchParamGroup. <i>UlPreallocationDataVolume</i>	Set this parameter based on site requirements.
Policies for controlling the number of UEs for preallocation	PRESCH_UE_NUM_CTRL_POL_SW option of the NRDUCellULSch.<i>UlPreallocPolEnhSwitch</i> parameter	Deselect this option. When this option is deselected, the maximum number of UEs for uplink preallocation is specified by the NRDUCellPusch.<i>PreschUeNumUpperLimit</i> parameter, irrespective of UE types.
Prescheduled UE Num Upper Limit	NRDUCellPusch.<i>PreschUeNumUpperLimit</i>	It is recommended that this parameter be set to 8 .

- QCI-based uplink smart preallocation

In QCI-based uplink smart preallocation, the gNodeB continuously schedules a UE for a certain period, but this continuous scheduling happens only if the gNodeB has downlink data to transmit. In contrast to QCI-based uplink basic preallocation (in which the gNodeB schedules a UE at regular intervals), QCI-based uplink smart preallocation can reduce uplink resource consumption.

QCI-specific SR Period Configuration

With this function, the SR period can be customized for the bearer intended for URLLC.

QCI-specific First-Packet Size Configuration for SR-based Scheduling

With this function, a different first-packet size in SR-based scheduling can be configured for the bearer intended for URLLC, reducing the delay in uplink packet transmission.

QCI-specific Scheduling Based on the Target IBLER

This function is classified into 'QCI-specific scheduling based on the uplink target IBLER' and 'QCI-specific scheduling based on the downlink target IBLER'. For a service using a QCI for which uplink and downlink target IBLERs are configured, the uplink and downlink IBLERs for the service can converge on the target IBLERs through outer loop link adaptation (OLLA). In response, the target IBLERs can be reduced to decrease the BLER and number of retransmissions, thereby reducing the air interface delay.

5.1.2 Network Analysis

5.1.2.1 Benefits

For details, see related descriptions in *Service-differentiated Experience Guarantee*.

5.1.2.2 Impacts

For details, see related descriptions in *Service-differentiated Experience Guarantee*.

5.1.3 Requirements

5.1.3.1 Licenses

For details, see related descriptions in *Service-differentiated Experience Guarantee*.

5.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.1.3.2.1 Prerequisite Functions

For details, see related descriptions in *Service-differentiated Experience Guarantee*.

5.1.3.2.2 Mutually Exclusive Functions

For details, see related descriptions in *Service-differentiated Experience Guarantee*.

5.1.3.2.3 Function Impacts

For details, see related descriptions in *Service-differentiated Experience Guarantee*.

5.1.3.3 Hardware

For details, see related descriptions in *Service-differentiated Experience Guarantee*.

5.1.3.4 Others

None

5.1.4 Operation and Maintenance

5.1.4.1 Data Configuration

5.1.4.1.1 Data Preparation

Table 5-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-1 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Low Latency Switch	NRDUCellAlgoSwitch.LowLatencySwitch	SERV_DIFF_LOW_LATENCY_SW	To enable service-differentiated low latency, select this option.
Low Latency Switch	NRDUCellAlgoSwitch.LowLatencySwitch	SERV_DIFF_POLICY_SW	To use a new policy for service differentiation, select this option.
Schedule Parameter Group ID	NRDUCellQciBearer.ScheduleParamGroupId	None	Set this parameter based on the network plan.
Schedule Parameter Group ID	gNBDUSchParamGroup.ScheduleParamGroupId	None	Set this parameter based on the network plan.
Schedule Option Switch	gNBDUSchParamGroup.ScheduleOptionSwitch	UL_PREALLOCATION_QCI_SW	To enable QCI-based uplink preallocation, select this option. QCI-based uplink preallocation and QCI-based uplink smart preallocation cannot be both enabled.
Schedule Option Switch	gNBDUSchParamGroup.ScheduleOptionSwitch	UL_IBLER_QCI_SW	To enable QCI-specific scheduling based on the uplink target IBLER, select this option.

Parameter Name	Parameter ID	Option	Setting Notes
Schedule Option Switch	gNBDUSchParamGroup.ScheduleOptionSwitch	DL_IBLER_QCI_SW	To enable QCI-specific scheduling based on the downlink target IBLER, select this option.
Schedule Option Switch	gNBDUSchParamGroup.ScheduleOptionSwitch	UL_SMART_PREALLOC_QCI_SW	To enable QCI-based uplink smart preallocation, select this option. QCI-based uplink smart preallocation and QCI-based uplink preallocation cannot be both enabled.
Schedule Option Switch	gNBDUSchParamGroup.ScheduleOptionSwitch	SR_PERIOD_QCI_SW	To enable the QCI-specific SR period configuration function, select this option.
Uplink Preallocation Interval	gNBDUSchParamGroup.UlPreallocationInterval	None	After QCI-based uplink preallocation is enabled, set this parameter based on the network plan.
Uplink Preallocation Data Volume	gNBDUSchParamGroup.UlPreallocationDataVolume	None	After QCI-based uplink preallocation is enabled, set this parameter based on the network plan.
UL Preallocation Policy Enhancement Switch	NRDUCellUISch.UlPreallocPolEnhSwitch	PRESCH_UE_NUM_CTRL_POL_SW	Deselect this option.
Prescheduled UE Num Upper Limit	NRDUCellPusch.PreschUeNumUpperLimit	None	It is recommended that this parameter be set to 8 .
Uplink Smart Preallocation Duration	gNBDUSchParamGroup.UlSmartPreallocationDuration	None	After QCI-based uplink smart preallocation is enabled, set this parameter based on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
SR Period	gNBDUSchParamGroup.SrPeriod	None	After QCI-specific SR period configuration is enabled, you are advised to set this parameter based on the actual service packet size. For example, if the service packet size is 500 bytes, you are advised to set this parameter to 4160 (in the unit of bit), considering 20 bytes of header overhead.
SR-based Scheduling First TB Size	gNBDUSchParamGroup.SrSchFirstTbSize	None	After the 'QCI-specific first-packet size configuration for SR-based scheduling' function is enabled, set this parameter based on the network plan.
Uplink Target IBLER	gNBDUSchParamGroup.UlTargetIbler	None	After QCI-specific scheduling based on the uplink target IBLER is enabled, set this parameter based on the network plan.
Downlink Target IBLER	gNBDUSchParamGroup.DlTargetIbler	None	After QCI-specific scheduling based on the downlink target IBLER is enabled, set this parameter based on the network plan.

5.1.4.1.2 Using MML Commands

Before using MML commands, refer to [5.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After

Modification" and "Caution" for the corresponding parameters in the parameter reference.

NOTICE

This function takes effect 5 to 10 minutes after the relevant parameter configurations in the **gNBDUSchParamGroup** and **NRDUCellQciBearer** MOs are modified.

Activation Command Examples

Basic configuration for service-differentiated low latency

```
//Turning on the switch for service-differentiated low latency
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, LowLatencySwitch=SERV_DIFF_LOW_LATENCY_SW-1;
//(Optional) Selecting SERV_DIFF_POLICY_SW
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, LowLatencySwitch=SERV_DIFF_POLICY_SW-1;
//Adding a gNodeB DU scheduling parameter group
ADD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1;
//Mapping the gNodeB DU scheduling parameter group to a bearer of a specific QCI
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=9, ScheduleParamGroupId=1;
```

QCI-based uplink preallocation (QCI-based uplink preallocation and QCI-based uplink smart preallocation cannot be enabled at the same time.)

- Enabling QCI-based uplink preallocation

```
//Selecting UL_PREALLOCATION_QCI_SW
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1,
ScheduleOptSwitch=UL_PREALLOCATION_QCI_SW-1;
//Modifying the interval and data volume for uplink preallocation
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, UIPreallocationInterval=5,
UIPreallocationDataVolume=100;
//Deselecting PRESCH_UE_NUM_CTRL_POL_SW
MOD NRDUCELLULSCH: NrDuCellId=1, UIPreallocPolEnhSwitch=PRESCH_UE_NUM_CTRL_POL_SW-0;
//Configuring the maximum number of UEs for preallocation
MOD NRDUCELLPUSCH: NrDuCellId=1, PreschUeNumUpperLimit=8;
```
- Enabling QCI-based uplink smart preallocation

```
//Selecting UL_SMART_PREALLOC_QCI_SW
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1,
ScheduleOptSwitch=UL_SMART_PREALLOC_QCI_SW-1;
//Modifying the interval and data volume for uplink preallocation and the duration for uplink smart preallocation
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, UIPreallocationInterval=5,
UIPreallocationDataVolume=100,UISmartPreallocationDur=250;
//Deselecting PRESCH_UE_NUM_CTRL_POL_SW
MOD NRDUCELLULSCH: NrDuCellId=1, UIPreallocPolEnhSwitch=PRESCH_UE_NUM_CTRL_POL_SW-0;
//Configuring the maximum number of UEs for preallocation
MOD NRDUCELLPUSCH: NrDuCellId=1, PreschUeNumUpperLimit=8;
```

QCI-specific SR period configuration

```
//Turning on the switch for QCI-specific SR period configuration
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, ScheduleOptSwitch=SR_PERIOD_QCI_SW-1;
//Modifying the SR period
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, SrPeriod=20SLOT;
```

QCI-specific first-packet size configuration for SR-based scheduling

```
//Modifying the size of the first packet of a specific QCI in SR-based scheduling
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, SrSchFirstTbSize=600;
```

QCI-specific scheduling based on the uplink target IBLER

```
//Turning on the switch for QCI-specific scheduling based on the uplink target IBLER  
MOD GNBUSCHPARAMGROUP: ScheduleParamGroupId=1, ScheduleOptSwitch=UL_IBLER_QCI_SW-1;  
//Changing the QCI-based uplink target IBLER to 10%  
MOD GNBUSCHPARAMGROUP: ScheduleParamGroupId=1, ULTargetIbler=10000;
```

QCI-specific scheduling based on the downlink target IBLER

```
//Turning on the switch for QCI-specific scheduling based on the downlink target IBLER  
MOD GNBUSCHPARAMGROUP: ScheduleParamGroupId=1, ScheduleOptSwitch=DL_IBLER_QCI_SW-1;  
//Changing the QCI-based downlink target IBLER to 10%  
MOD GNBUSCHPARAMGROUP: ScheduleParamGroupId=1, DLTargetIbler=10000;  
//Mapping the gNodeB DU scheduling parameter group to a bearer of a specific QCI  
MOD NRDUCELLQCI BEARER: NrDuCellId=1, Qci=9, ScheduleParamGroupId=1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling QCI-based uplink preallocation  
MOD GNBUSCHPARAMGROUP: ScheduleParamGroupId=1,  
ScheduleOptSwitch=UL_PREALLOCATION_QCI_SW-0;  
  
//Disabling QCI-based uplink smart preallocation  
MOD GNBUSCHPARAMGROUP: ScheduleParamGroupId=1,  
ScheduleOptSwitch=UL_SMART_PREALLOC_QCI_SW-0;  
  
//(Optional) Disabling the new policy for service differentiation  
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, LowLatencySwitch=SERV_DIFF_POLICY_SW-0;  
  
//Disabling QCI-specific SR period configuration  
MOD GNBUSCHPARAMGROUP: ScheduleParamGroupId=1, ScheduleOptSwitch=SR_PERIOD_QCI_SW-0;  
  
//Disabling the QCI-specific first-packet size configuration for SR-based scheduling function  
MOD GNBUSCHPARAMGROUP: ScheduleParamGroupId=1, SrSchFirstTbSize=0;  
  
//Disabling QCI-specific scheduling based on the uplink target IBLER  
MOD GNBUSCHPARAMGROUP: ScheduleParamGroupId=1, ScheduleOptSwitch=UL_IBLER_QCI_SW-0;  
  
//Disabling QCI-specific scheduling based on the downlink target IBLER  
MOD GNBUSCHPARAMGROUP: ScheduleParamGroupId=1, ScheduleOptSwitch=DL_IBLER_QCI_SW-0;  
  
//Removing the mapping between the gNodeB DU scheduling parameter group and the bearer of the  
specific QCI  
MOD NRDUCELLQCI BEARER: NrDuCellId=1, Qci=9, ScheduleParamGroupId=255;  
  
//Removing the gNodeB DU scheduling parameter group  
RMV GNBUSCHPARAMGROUP: ScheduleParamGroupId=1;  
  
//Turning off the switch for service-differentiated low latency  
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, LowLatencySwitch=SERV_DIFF_LOW_LATENCY_SW-0;
```

5.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.1.4.2 Activation Verification

For details, see related descriptions in *Service-differentiated Experience Guarantee*.

5.1.4.3 Network Monitoring

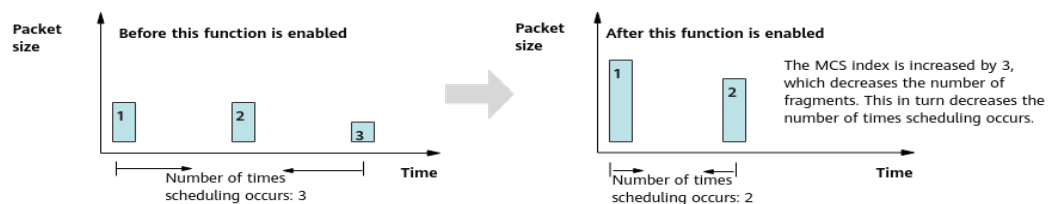
For details, see related descriptions in *Service-differentiated Experience Guarantee*.

5.2 Uplink Segmentation-Free Scheduling

5.2.1 Principles

The uplink segmentation-free scheduling function is introduced to reduce the delay caused by the fragmentation process in uplink scheduling. This function **increases the uplink MCS index by a fixed value of 3 for UEs with delay-sensitive factors configured to improve the uplink spectral efficiency, which in turn reduces the number of uplink segments or prevents uplink segmentation.** The configuration of delay-sensitive factors for UEs is indicated by the setting of the `gNBDUSchParamGroup.CycleTime` parameter to a value other than `NOT_CONFIG`. [Figure 5-1](#) illustrates this function. This function is controlled by the `UL_SEGMENT_FREE_SCH_SW` option of the `NRDUCellUISch.UISchAlgoSwitch` parameter.

Figure 5-1 Uplink segmentation-free scheduling



For a cell serving multiple UEs, it is recommended that this function be enabled when the average uplink MCS index drops below 5 and the number of RBs scheduled in the uplink is less than 50.

5.2.2 Network Analysis

5.2.2.1 Benefits

Uplink segmentation-free scheduling increases the uplink MCS index by a fixed value of 3 for a UE configured with the delay-sensitive factor. This reduces the number of uplink segments or prevents uplink segmentation, which in turn reduces delay.

Conditions for Optimal Gains

For details, see [Conditions for Optimal Gains](#).

5.2.2.2 Impacts

The impacts between this function and other functions are described in [5.2.3.2.3 Function Impacts](#).

There are no impacts on the network.

5.2.3 Requirements

5.2.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-051206	URLLC Introduction	NR0SURLLCP00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.2.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Scheduling based on delay-sensitive requirements	gNBDUSchParamGroup.CycleTime	<i>URLLC</i>	Uplink segmentation-free scheduling takes effect only for UEs for which gNBDUSchParamGroup.CycleTime is set to a value other than NOT_CONFIG .

5.2.3.2.2 Mutually Exclusive Functions

None

5.2.3.2.3 Function Impacts

None

5.2.3.3 Hardware

For details, see [4.3.3 Hardware](#).

5.2.3.4 Others

None

5.2.4 Operation and Maintenance

5.2.4.1 Data Configuration

5.2.4.1.1 Data Preparation

[Table 5-2](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-2 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
UL Scheduling Algorithm Switch	NRDUCellUISch.UISchAlgoSwitch	UL_SEGMENT_FREE_SCH_SW	To enable uplink segmentation-free scheduling, select this option. It is recommended that this function be enabled when the average uplink MCS index drops below 5 and the number of RBs scheduled in the uplink is less than 50 in multi-UE scenarios.

5.2.4.1.2 Using MML Commands

Before using MML commands, refer to [5.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling uplink segmentation-free scheduling  
MOD NRDUCELLULSCH: NrDuCellId=1, UISchAlgoSwitch=UL_SEGMENT_FREE_SCH_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling uplink segmentation-free scheduling  
MOD NRDUCELLULSCH: NrDuCellId=1, UISchAlgoSwitch=UL_SEGMENT_FREE_SCH_SW-0;
```

5.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.2.4.2 Activation Verification

Run the **LST NRDUCELLULSCH** command to query the **UL Segmentation-Free Scheduling Sw** setting of the **UL Scheduling Algorithm Switch** parameter to see whether the corresponding function has been enabled.

5.2.4.3 Network Monitoring

For details, see [4.4.3 Network Monitoring](#).

6 Reliability Improvement

This section describes the functions that are oriented toward reliability improvement.

6.1 Data Channel Reliability Optimization

6.1.1 Principles

Data channel reliability is guaranteed by a number of functions, including resource reservation, OLLA algorithm optimization, CQI table 3, MCS index table 3, PDCP out-of-order delivery, service-differentiated low latency, HARQ retransmission optimization, uplink differentiated power control, and rank configuration for low latency and high reliability.

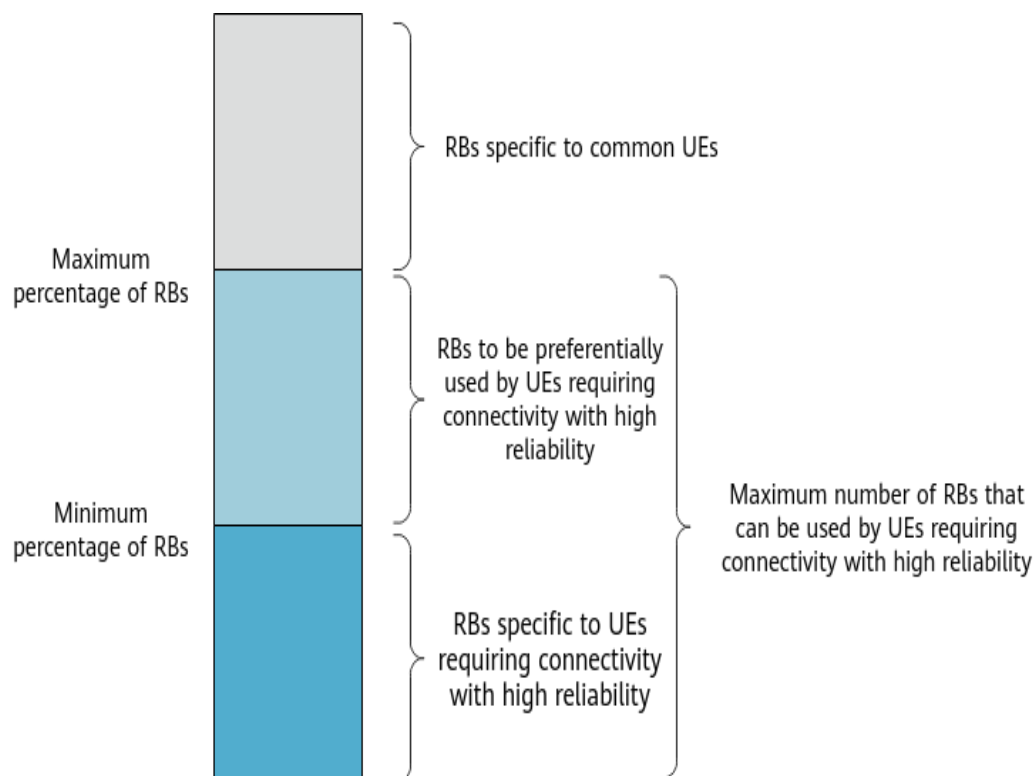
Resource Reservation

Resource reservation ensures that PDSCH and PUSCH channel resources are preferentially used for services requiring connectivity with high reliability. High-reliability-oriented resource reservation requires that the scheduling type be set to high-reliability-oriented scheduling (with the **gNBDUMacParamGroup.SchedulingType** parameter set to **HIGH_RELIABILITY_SCH**). Two resource reservation modes—slicing-based resource reservation and independent resource reservation in non-slicing scenarios—are available, but they cannot be both enabled.

- Slicing-based resource reservation: **If slices are configured, resources are reserved for slice groups and services requiring connectivity with high reliability are configured to run on the slices, thereby guaranteeing the provision of such services.** This type of reservation requires that the **NRDUCellRes.ResUsageObject** parameter be set to **NETWORK_SLICE_GROUP**. This configuration indicates that the given template is used for slice objects. For details, see descriptions of NR DU cell resource management between network slices in *Network Slicing*.
- Independent resource reservation in non-slicing scenarios: **If no slices are configured, fixed resources are reserved for services requiring connectivity with high reliability to guarantee the provision of such services.** In this type of resource reservation, the following items related to the PUSCH and

PDSCH must be configured to define the range of RB resources available to UEs requiring connectivity with high reliability and that available to common UEs: minimum percentages of RBs, frequency-domain start position, and maximum percentage of RBs. **Figure 6-1** shows an example of this type of resource reservation.

Figure 6-1 RB resource configuration in non-slicing scenarios



Independent resource reservation in non-slicing scenarios is controlled by the **RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellAlgoSwitch.HighReliabilitySwitch** parameter. In addition, this type of reservation requires that PUSCH and PDSCH resource reservation parameters be configured. **Table 6-1** and **Table 6-2** list these parameters.

Table 6-1 PUSCH resource reservation parameters

Parameter	Description
NRDUCellRes.ResUsageObject	Indicates a resource usage object. Set this parameter to HIGH_RELIABILITY . This configuration indicates that the given template is used for high reliability.
NRDUCellRes.ResStrategyType	Indicates an RB resource control policy. Set this parameter to TTI_ONLY .
NRDUCellRes.UIRbMinRatio	Indicates the minimum percentage of PUSCH RBs available for UEs requiring connectivity with high reliability. The RBs within the RB quota determined by this parameter are exclusively used by such UEs.

Parameter	Description
NRDUCellRes.UIRbMaxRatio	Indicates the maximum percentage of PUSCH RBs available for UEs requiring connectivity with high reliability. The RBs within the RB quota determined by this parameter are preferentially used by such UEs.
NRDUCellRes.UIMinRbStartPos	Indicates the start position of the RB quota determined by the NRDUCellRes.UIRbMinRatio parameter for UEs requiring connectivity with high reliability.

Table 6-2 PDSCH resource reservation parameters

Parameter	Description
NRDUCellRes.ResUsageObject	Indicates a resource usage object. Set this parameter to HIGH_RELIABILITY . This configuration indicates that the given template is used for high reliability.
NRDUCellRes.ResStrategyType	Indicates an RB resource control policy. Set this parameter to TTI_ONLY .
NRDUCellRes.DIRbMinRatio	Indicates the minimum percentage of PDSCH RBs available for UEs requiring connectivity with high reliability. The RBs within the RB quota determined by this parameter are exclusively used by such UEs.
NRDUCellRes.DIRbMaxRatio	Indicates the maximum percentage of PDSCH RBs available for UEs requiring connectivity with high reliability. The RBs within the RB quota determined by this parameter are preferentially used by such UEs.
NRDUCellRes.DIMinRbStartPos	Indicates the start position of the RB quota determined by the NRDUCellRes.DIRbMinRatio parameter for UEs requiring connectivity with high reliability.

NOTE

When RBG-granularity scheduling is used, the maximum number of available RBs must not be less than the number of RBs in one RBG for both UEs requiring connectivity with high reliability and other types of UEs. In this case, the actual number of RBs that can be scheduled will be less than the configured maximum number of RBs. The number of RBs in an RBG varies depending on the bandwidth. For details, see "Table 5.3.2-1: Transmission bandwidth configuration NRB for FR1" in section 5.3.2 "Transmission bandwidth configuration" of 3GPP TS 38.104 V15.5.0.

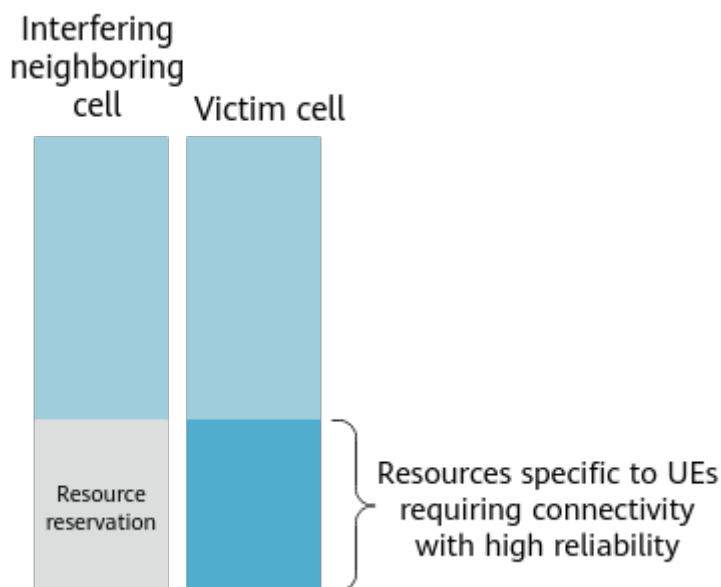
- For details about uplink RBGs, see "Table 6.1.2.2.1-1: Nominal RBG size P" in section 6.1.2.2.1 "Uplink resource allocation type 0" of 3GPP TS 38.214 V15.4.0.
- For details about downlink RBGs, see "Table 5.1.2.2.1-1: Nominal RBG size P" in section 5.1.2.2.1 "Downlink resource allocation type 0" of 3GPP TS 38.214 V15.4.0.

When either of the two resource reservation modes is used in a cell to ensure that services requiring connectivity with high reliability are provided with dedicated resources, the cell may be affected by interference from a neighboring cell that is heavily loaded in the uplink (for example, when the PRB usage in the neighboring cell exceeds 50%). To reduce intra-frequency interference from the neighboring cell to the high-reliability cell, it is necessary to avoid situations where common UEs in the neighboring cell and UEs requiring connectivity with high reliability in the high-reliability cell are scheduled on the same RB positions. Accordingly, it is recommended to configure related parameters for the high-reliability cell and interfering neighboring cell to reserve fixed resources, as described below:

- There is a single high-reliability-oriented victim cell: The frequency-domain positions for high-reliability-oriented resource reservation are aligned between the neighboring cell and high-reliability-oriented victim cell, as shown in [Figure 6-2](#). This alignment prevents common UEs in the neighboring cell from being scheduled on the reserved frequency-domain resource positions, thereby reducing interference from common UEs in the neighboring cell to UEs requiring connectivity with high reliability.

Parameter	Interfering Neighboring Cell	High-Reliability-Oriented Victim Cell (Quantity: One)
NRDUCellRes.UlMinRbStartPos	Set this parameter to 0 . This configuration indicates that the fixed resources are allocated starting from the PRB directly following the PUCCH end PRB.	
NRDUCellRes.DlMinRbStartPos	Set this parameter to 0 . This configuration indicates that the fixed resources are allocated starting from RB0.	
NRDUCellRes.UlRbMinRatio	Set this parameter to the same value for the neighboring cell and high-reliability cell.	
NRDUCellRes.UlRbMaxRatio	Set this parameter to the same value for the neighboring cell and high-reliability cell.	
NRDUCellRes.DlRbMinRatio	Set this parameter to the same value for the neighboring cell and high-reliability cell.	
NRDUCellRes.DlRbMaxRatio	Set this parameter to the same value for the neighboring cell and high-reliability cell.	
INIT_BWP_FULL_BW_S option of the NRDUCellAlgoSwitch.BwpConfigPolicy-Switch parameter	No requirement	Select this option.

Figure 6-2 Resource reservation in the presence of interference

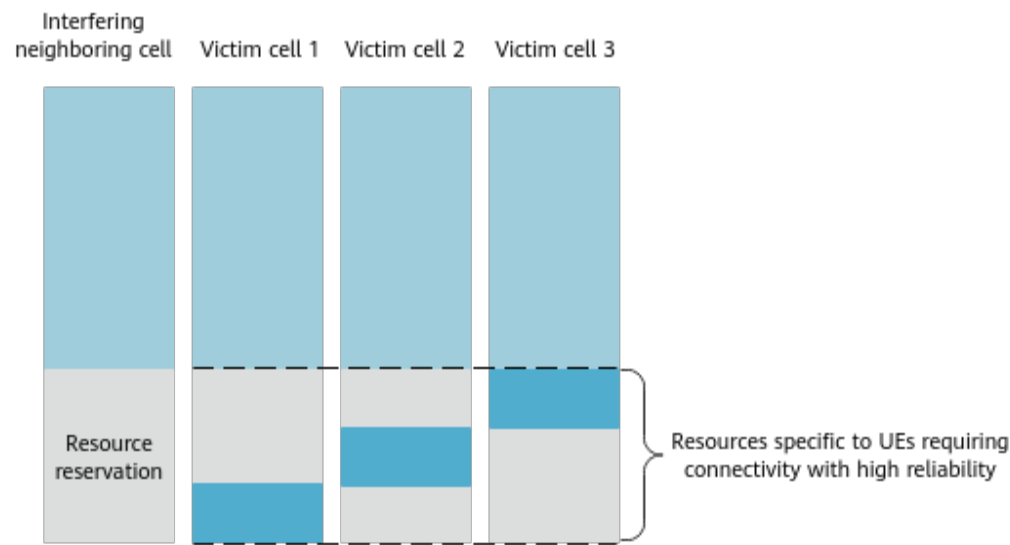


- There are multiple high-reliability-oriented victim cells: To reduce interference from neighboring cells, the frequency-domain positions for high-reliability-oriented resource reservation are aligned between high-reliability-oriented victim cells and neighboring cells. In addition, to reduce mutual interference among high-reliability cells, interference randomization is applied to uplink scheduling within the reserved fixed resources, as shown in [Figure 6-3](#).

Parameter	Interfering Neighboring Cell	High-Reliability-Oriented Victim Cell (Quantity: More than One)
NRDUCellRes.UlMinRbStartPos	Set this parameter to 0 . This configuration indicates that the fixed resources are allocated starting from the PRB directly following the PUCCH end PRB.	
NRDUCellRes.DlMinRbStartPos	Set this parameter to 0 . This configuration indicates that the fixed resources are allocated starting from RB0.	
NRDUCellRes.UlRbMinRatio	Set this parameter to the same value for all cells, and set NRDUCellRes.UlRbMinRatio and NRDUCellRes.UlRbMaxRatio to the same value.	
NRDUCellRes.UlRbMaxRatio	Set this parameter to the same value for all cells, and set NRDUCellRes.UlRbMinRatio and NRDUCellRes.UlRbMaxRatio to the same value.	
NRDUCellRes.DlRbMinRatio	Set this parameter to the same value for all cells, and set NRDUCellRes.DlRbMinRatio and NRDUCellRes.DlRbMaxRatio to the same value.	
NRDUCellRes.DlRbMaxRatio	Set this parameter to the same value for all cells, and set NRDUCellRes.DlRbMinRatio and NRDUCellRes.DlRbMaxRatio to the same value.	

Parameter	Interfering Neighboring Cell	High-Reliability-Oriented Victim Cell (Quantity: More than One)
INIT_BWP_FULL_BW_SW option of the NRDUCellAlgoSwitch.BwpConfigPolicy-Switch parameter	Select this option. With this setting, the gNodeB configures a full-bandwidth initial BWP for all UEs.	
FIX_RES_UL_INTRF_R ANDOM_SCH_SW option of the NRDUCellRes.SchedulingPolicySwitch parameter	No requirement	Select this option. With this setting, interference randomization is applied to uplink scheduling within the reserved fixed resources.

Figure 6-3 Interference randomization



OLLA Algorithm Optimization

In scenarios with significant signal fluctuations, the MCS varies widely and the BER fails to converge. To improve reliability for UEs requiring connectivity with low latency and high reliability under such conditions, the OLLA algorithm optimization function is introduced. In this function, the allowable MCS index range is establish for the PDSCH and PUSCH by configuring the maximum and minimum MCS indexes, reducing the BER when signal fluctuations occur. [Figure 6-4](#) illustrates how this function works.

Figure 6-4 Optimization of the MCS index range

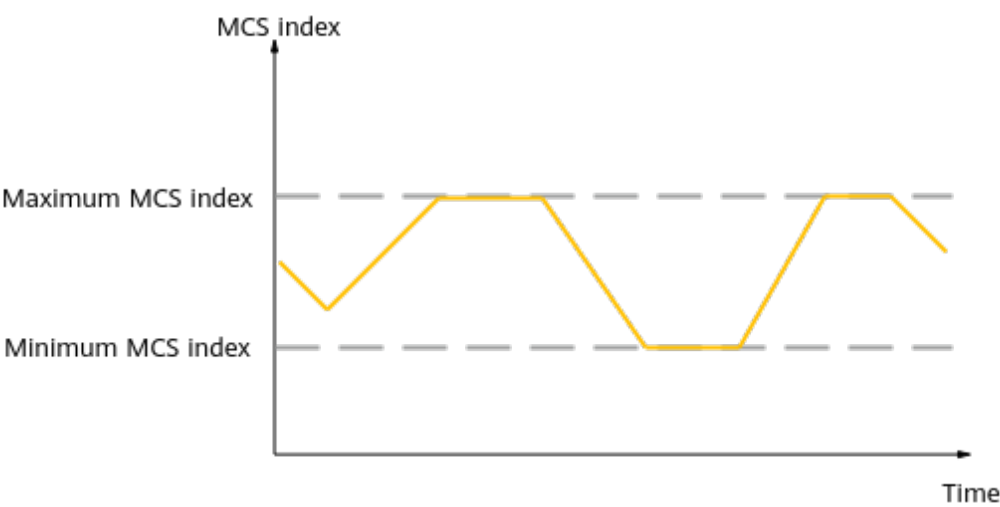


Table 6-3 and **Table 6-4** list the parameters that must be configured for this function to work.

Table 6-3 Parameters related to OLLA algorithm optimization for the PDSCH

Parameter	Description
gNBDUSchParamGroup.DlInitialMcsAdjValue	Indicates the initial adjustment value of the MCS for downlink transmission.
gNBDUSchParamGroup.DlAmcAdjustStep	Indicates the downlink AMC adjustment step.
gNBDUSchParamGroup.DlMaxMcs	Indicates the maximum downlink MCS index.
gNBDUSchParamGroup.DlMinMcs	Indicates the minimum downlink MCS index.

Table 6-4 Parameters related to OLLA algorithm optimization for the PUSCH

Parameter	Description
gNBDUSchParamGroup.UlInitialMcsAdjValue	Indicates the initial adjustment value of the MCS for uplink transmission.
gNBDUSchParamGroup.UlAmcAdjustStep	Indicates the uplink AMC adjustment step.
gNBDUSchParamGroup.UlMaxMcs	Indicates the maximum uplink MCS index.
gNBDUSchParamGroup.UlMinMcs	Indicates the minimum uplink MCS index.

NOTE

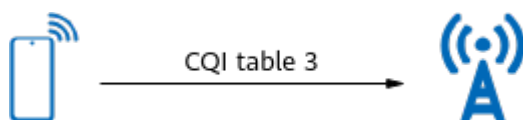
If a QCI is not defined in the **NRDUCellQciBearer** MO, the preceding parameters for the QCI use their default values rather than the manually configured values.

CQI Table 3

In digital communications, higher-order modulation leads to higher spectrum utilization but also a higher BER. To ensure the reliability of services requiring connectivity with high reliability, 3GPP TS 38.214 defines CQI tables tailored to 5G services requiring connectivity with a reliability target of 99.999%. One such example is CQI table 3. **This table defines lower code rates and spectral efficiency than other CQI tables for the same CQI index. Using this table helps improve data transmission reliability.** Against this background, the CQI table 3 function is introduced. This function enables UEs to report measurement results using CQI table 3, thereby reducing the BLER of services requiring connectivity with high reliability. [Figure 6-5](#) illustrates this function. For details about CQI table 3, see section 5.2.2.1 "Channel quality indicator (CQI)" in 3GPP TS 38.214 V17.6.0.

The **CQI_TABLE3_MEAS_SW** option of the **NRDUCellQciBearer.ServiceDiffAlgoSwitch** parameter determines whether to use CQI table 3.

Figure 6-5 CQI table 3

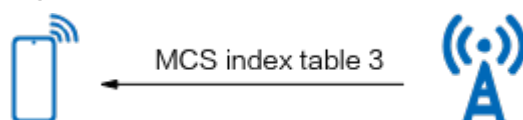


MCS Index Table 3

In digital communications, higher-order modulation leads to higher spectrum utilization but also a higher BER. To ensure the reliability of services requiring connectivity with high reliability, 3GPP TS 38.214 defines MCS index tables tailored to 5G services requiring connectivity with a reliability target of 99.999%. One such example is MCS index table 3. **This table defines lower modulation orders and target code rates than other MCS index tables for the same MCS index. Using this table helps improve data transmission reliability.** Against this background, the MCS index table 3 function is introduced. After this function is enabled, the base station uses MCS index table 3 (lower code rates) in downlink scheduling for UEs requiring connectivity with low latency and high reliability, thereby reducing the BLER of services requiring connectivity with high reliability for these UEs. [Figure 6-6](#) illustrates this function. For details about this MCS index table, see section 5.1.3.1 "Modulation order and target code rate determination" in 3GPP TS 38.214 V17.6.0.

The **MCS_TABLE3_SW** option of the **NRDUCellQciBearer.ServiceDiffAlgoSwitch** parameter determines whether to use MCS index table 3.

Figure 6-6 MCS index table 3

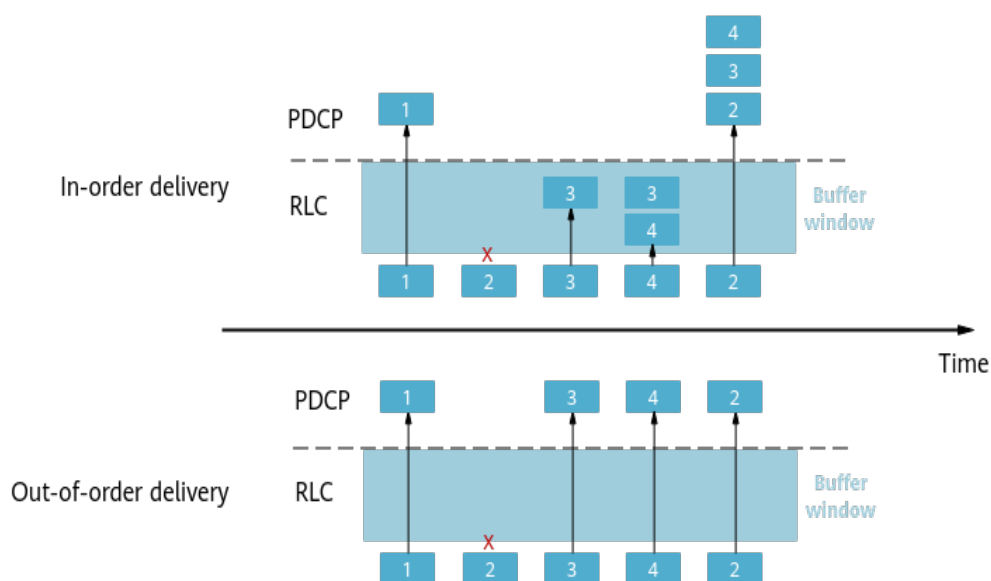


PDCP Out-of-Order Delivery

To reduce the processing time at the receiver, the PDCP out-of-order delivery function is introduced. **Figure 6-7** shows how this function works. If the receiver supports PDCP out-of-order delivery, PDCP reordering is not initiated in the event of packet disorder. **This means that out-of-order packets will be transmitted to the upper layer for processing, thereby reducing delay.** This function includes two forms of out-of-order delivery: one at the base station level and the other at the UE level, differing in receivers. Details are as follows:

- PDCP out-of-order delivery at the base station level: This function is enabled when the **gNBpdcpparamGroup.GnbPdcppOutOfOrderDeliveryInd** parameter is set to **TRUE_WITHOUT_UE_CAPB** or **TRUE_WITH_UE_CAPB**. After a UE sends data packets over a bearer to the base station, the base station's bearer-specific PDCP entity directly forwards the received data packets to the core network without reordering them.
- PDCP out-of-order delivery at the UE level: This function is enabled when the **gNBpdcpparamGroup.UePdcppOutOfOrderDeliveryInd** parameter is set to **TRUE**. If this function is enabled for a UE that supports PDCP out-of-order delivery, the PDCP entity of the UE directly forwards received data packets to the application layer without reordering them.

Figure 6-7 PDCP out-of-order delivery



Service-Differentiated Low Latency

Service-differentiated low latency can be enabled to reduce the air interface latency for UEs requiring connectivity with high reliability. For details about this function, see [5.1 Service-Differentiated Low Latency](#). It is recommended that the **gNBdusParamGroup.UlPreallocationDataVolume** parameter be set to twice the service packet size.

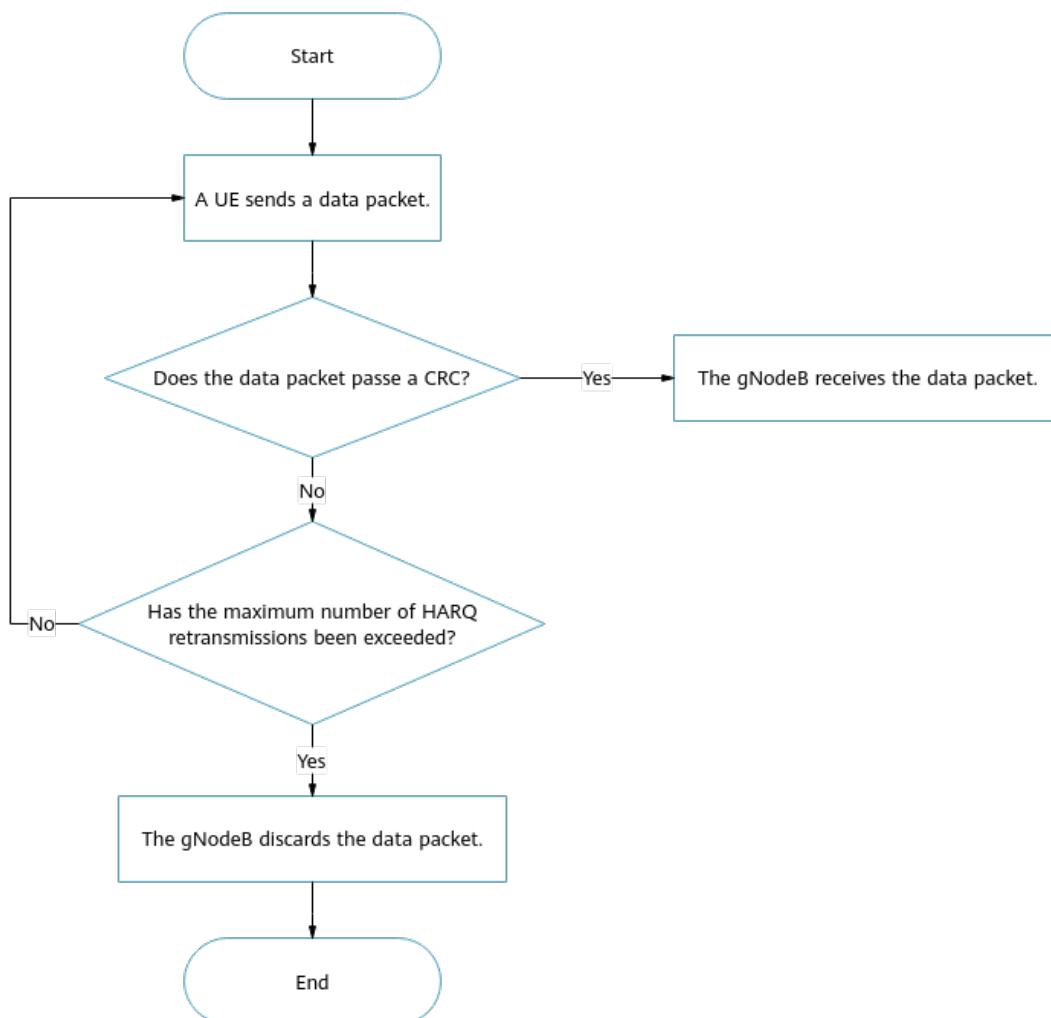
HARQ Retransmission Optimization

This function limits the number of HARQ retransmissions. A data packet will be discarded if it has been retransmitted more times than the preset upper limit. This

avoids unnecessary resource consumption and ensures that other data packets can be transmitted and received efficiently, thereby shortening the waiting time.

Figure 6-8 shows the HARQ retransmission optimization process.

Figure 6-8 HARQ retransmission optimization



HARQ retransmission optimization requires the following parameters to be configured:

- **gNBDUSchParamGroup.UIMaxHarqRetransNum** and **gNBDUSchParamGroup.DIMaxHarqRetransNum**: specify the maximum number of HARQ retransmissions for the uplink and downlink, respectively.
- **gNBPdcpParamGroup.GnbPdcpOutOfOrderDeliveryInd**: specifies whether to use PDCP out-of-order delivery on the gNodeB and is set to **TRUE_WITHOUT_UE_CAPB** or **TRUE_WITH_UE_CAPB**.
- **NRCellQciBearer.RlcMode**: specifies the RLC transmission mode.
 - When the transport layer uses the TCP protocol, it is recommended that the RLC transmission mode be set to **AM** and the following parameters be reconfigured based on the maximum numbers of HARQ retransmissions configured for the uplink and downlink:

Parameter	Description
gNBRLcParamGroup.gNB PollingPduRetransTimer	Indicates the length of the polling PDU retransmission timer for the gNodeB.
gNBRLcParamGroup.UeP ollingPduRetransTimer	Indicates the length of the polling PDU retransmission timer for a UE.
gNBRLcParamGroup.gNB RlcReassemblyTimer	Indicates the length of the gNodeB-specific timer for RLC reassembly.
gNBRLcParamGroup.UeRl cReassemblyTimer	Indicates the length of the UE-specific timer for RLC reassembly.
gNBRLcParamGroup.gNB AmStatusRptProhibitTmr	Indicates the length of the gNodeB-specific timer for prohibiting status reporting in acknowledged mode (AM).
gNBRLcParamGroup.UeA mStatusRptProhibitTmr	Indicates the length of the UE-specific timer for prohibiting status reporting in AM.

NOTE

In TDD high-frequency networking, if the **gNBRLcParamGroup.RadioBearerType** parameter is set to **SRB1_FR2**, the settings of the **gNBRLcParamGroup.gNBPollingPduRetransTimer** and **gNBRLcParamGroup.UePollingPduRetransTimer** parameters are not applied. Instead, the length of the polling PDU retransmission timer for the gNodeB and that for UE are adaptively generated.

If a QCI is not defined in the **NRDUCellQciBearer** MO, the maximum number of downlink HARQ retransmissions for the QCI is determined by the **NRDUCellPdsch.MaxDlHarqTransNum** parameter rather than the **gNBDUSchParamGroup.DlMaxHarqRetransNum** parameter.

- When the transport layer uses the UDP protocol or a packet redundancy mechanism is employed, set the RLC transmission mode to **UM**.

Uplink Differentiated Power Control

To distinguish the PUSCH and SRS transmit power of common UEs from that of UEs requiring connectivity with low latency and high reliability, uplink differentiated power control is introduced. In this function, the **NRDUCellSrs.UrlcPoOffset** parameter is configured to specify the PUSCH and SRS PO offset for UEs requiring connectivity with low latency and high reliability relative to that of common UEs. Setting this parameter to a non-zero value increases the uplink transmit power of UEs requiring connectivity with low latency and high reliability, thereby improving the uplink reliability for URLLC services.

Rank Configuration for Low Latency and High Reliability

The data channel resource consumption is high if both uplink and downlink transmissions are fixed at rank 1 for UEs requiring connectivity with low latency and high reliability. To reduce data channel resource consumption, the rank

configuration for low latency and high reliability function is introduced. This function allows the maximum supported rank for both uplink and downlink transmissions to be increased to rank 2 for UEs requiring connectivity with low latency and high reliability. This function is controlled by the **gNBDUSchParamGroup.UrlcMaxRank** parameter, as described below:

- If this parameter is set to **NOT_CONFIG**, the maximum rank for uplink and downlink transmissions is not configured for UEs requiring connectivity with low latency and high reliability. In this case, both uplink and downlink transmissions are fixed at rank 1 for these UEs.
- If this parameter is set to **RANK1**, the maximum rank for uplink and downlink transmissions is rank 1 for UEs requiring connectivity with low latency and high reliability.
- If this parameter is set to **RANK2**, the maximum rank for uplink and downlink transmissions is rank 2 for UEs requiring connectivity with low latency and high reliability.

This function takes effect only on UEs requiring connectivity with low latency and high reliability. **RANK2** is recommended only in scenarios where frequency-domain resources are insufficient. With this configuration, data channel resource consumption is reduced due to spatial multiplexing, thereby increasing the number of such UEs that can be supported.

NOTE

If a QCI is not defined in the **NRDUCellQciBearer** MO, the **gNBDUSchParamGroup.UrlcMaxRank** parameter for the QCI uses the value **NOT_CONFIG**.

6.1.2 Network Analysis

6.1.2.1 Benefits

The following table lists the benefits of data channel reliability optimization.

Function	Benefit
Resource reservation	• It ensures that data channel resources are available for services requiring connectivity with low latency and high reliability. • It enables the positions of resources to be adjusted for UEs requiring connectivity with low latency and high reliability, which prevents potential interference that can occur in some frequency bands.
OLLA algorithm optimization	It reduces the BLER in the event of signal fluctuation.
CQI table 3	It reduces the BLER of services requiring connectivity with high reliability.
MCS index table 3	It reduces the BLER of services requiring connectivity with high reliability.

Function	Benefit
PDCP out-of-order delivery	It reduces the delay caused by data packet disorder.
HARQ retransmission optimization	It allows for flexible configuration of the number of HARQ retransmissions, thereby reducing the retransmission delay.
Uplink differentiated power control	It improves the accuracy of uplink SINR measurement, thereby increasing uplink reliability.
Rank configuration for low latency and high reliability	It is recommended that the gNBDUSchParamGroup.UrlcMaxRank parameter be set to RANK2 in scenarios where RB resources are insufficient. An example of such a scenario is special WAN. After rank 2 takes effect, it increases throughput for UEs requiring connectivity with low latency and high reliability and reduces consumption of RB resources. In other scenarios, it is recommended that this parameter be set to NOT_CONFIG or RANK1 to ensure reliability for UEs requiring connectivity with low latency and high reliability.

Conditions for Optimal Gains

For details, see [Conditions for Optimal Gains](#).

6.1.2.2 Impacts

The impacts between this function and other functions are described in [6.1.3.2.3 Function Impacts](#).

Functions related to data channel reliability optimization cause a high consumption of uplink and downlink resources due to high-reliability-oriented scheduling being applied to UEs requiring connectivity with low latency and high reliability. This reduces the resources available to other UEs.

Function	Impact on the Network
Resource reservation	No impact
OLLA algorithm optimization	No impact
CQI table 3	No impact
MCS index table 3	No impact

Function	Impact on the Network
PDCP out-of-order delivery	Unknown risks related to compatibility with the application layer may occur if both of the following conditions are met: (1) The base stations and UEs are configured to support PDCP out-of-order delivery; (2) The application layer is unable to effectively process out-of-order data packets or lacks the capability to process such packets.
HARQ retransmission optimization	No impact
Uplink differentiated power control	No impact
Rank configuration for low latency and high reliability	When the gNBDUSchParamGroup.UrlcMaxRank parameter is set to RANK2 , the throughput and bit error rate increase for UEs requiring connectivity with low latency and high reliability.

6.1.3 Requirements

6.1.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-051206	URLLC Introduction	NR0SURLLCP00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

6.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.1.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Identification of UEs requiring connectivity with low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter gNBDUMacParamGroup.SchedulingType set to HIGH_RELIABILITY_SCH gNBQciBearer.ServiceType set to URLLC_HIGH_RELIABILITY_SERVICE	<i>URLLC</i>	All sub-functions of data channel reliability optimization depend on identification of UEs requiring connectivity with low latency and high reliability.
Low-frequency TDD	Service-differentiated low latency	SERV_DIFF_LOW_LATENCY_SW option of the NRDUCellAlgoSwitch.LowLatencySwitch parameter	<i>Service-differentiated Experience Guarantee</i>	None
Low-frequency TDD	Uplink Min RB Start Position	NRDUCellRes.UlMinRbStartPos	<i>URLLC</i>	When the switch for uplink interference randomization-based scheduling within fixed resources is turned on, Uplink Min RB Start Position must be set to 0 .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink RB Max Ratio and Uplink RB Min Ratio	NRDUCellRes.UlRbMaxRatio and NRDUCellRes.UlRbMinRatio	URLLC	When the switch for uplink interference randomization-based scheduling within fixed resources is turned on, Uplink RB Max Ratio and Uplink RB Min Ratio must be set to the same value. In TDD, the NRDUCellRes.UlRbMinRatio parameter must be set to a value less than or equal to 10 if all the following conditions are met: (1) The REDCAP_SW option of the NRDUCellFeatureSw.UeCapbBasedFunSw parameter is selected; (2) The NRDUCellRes.ResUsageObject parameter is set to HIGH_RELIABILITY; (3) The NRDUCellRes.UlMinRbStartPos parameter is set to a value other than 65535.
Low-frequency TDD	Downlink Min RB Start Position and Downlink RB Min Ratio	NRDUCellRes.DlMinRbStartPos and NRDUCellRes.DlRbMinRatio	URLLC	The NRDUCellRes.DlRbMinRatio parameter must be set to a value less than or equal to 10 if all the following conditions are met: (1) The REDCAP_SW option of the NRDUCellFeatureSw.UeCapbBasedFunSw parameter is selected; (2) The NRDUCellRes.ResUsageObject parameter is set to HIGH_RELIABILITY; (3) The NRDUCellRes.DlMinRbStartPos parameter is set to a value other than 65535.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink Grant-Free Transform Precoder	gNBDUSchParamGroup.UlGfTransformPrecoder	None	When UL Grant-free is enabled, the following rule applies: If the gNBDUSchParamGroup.UllcMaxRank parameter is set to RANK2 , only the CP-OFDM waveform can be configured for UL Grant-free.

6.1.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NR DU cell resource management between network slices	RB_DYNAMIC_CONTROL_SW option of the NRDUCellAlgoSwitch.NetworkSliceAlgoSwitch parameter	<i>Network Slicing</i>	When network slicing is enabled, dynamic RB control cannot be enabled, that is, the RB_DYNAMIC_CONTROL_SW option of the NRDUCellAlgoSwitch.HighReliability-Switch parameter cannot be selected.
Low-frequency TDD	User PRB control for FWA	PRB_CTRL_SW option of the NRDUCellAlgoSwitch.FwaAlgoSwitch parameter	<i>FWA</i>	When user PRB control for FWA is enabled, dynamic RB control (controlled by the RB_DYNAMIC_CONTROL_SW option of the NRDUCellAlgoSwitch.HighReliability-Switch parameter) cannot be enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RFSP-based RB resource control	RB_DYNAMIC_CONTROL_SW option of the NRDUCellCommonSw.RfspAlgoSwitch parameter	<i>Resource Control in NSA Networking</i>	When RFSP-based RB resource control is enabled, dynamic RB control cannot be enabled, that is, the RB_DYNAMIC_CONTROL_SW option of the NRDUCellAlgoSwitch.HighReliability-Switch parameter cannot be selected.
Low-frequency TDD	Percentage-based dynamic RB sharing among operators	RB_DYNAMIC_SHARING_SW option of the NRDUCellAlgoSwitch.RanSharingAlgoSwitch parameter	<i>Multi-Operator Sharing</i>	When percentage-based dynamic RB sharing among operators is enabled, dynamic RB control cannot be enabled, that is, the RB_DYNAMIC_CONTROL_SW option of the NRDUCellAlgoSwitch.HighReliability-Switch parameter cannot be selected.
Low-frequency TDD	Operator-specific limitation on the maximum number of available RBs	MAX_RB_LIMIT_SW option of the NRDUCellAlgoSwitch.RanSharingAlgoSwitch parameter	<i>Multi-Operator Sharing</i>	When operator-specific limitation on the maximum number of available RBs is enabled, dynamic RB control cannot be enabled, that is, the RB_DYNAMIC_CONTROL_SW option of the NRDUCellAlgoSwitch.HighReliability-Switch parameter cannot be selected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Operator-specific guarantee on the minimum number of available RBs	MIN_RB_GUARANTEE_SW option of the NRDUCellAlgoSwitch.RanSharingAlgoSwitch parameter	<i>Multi-Operator Sharing</i>	When operator-specific guarantee on the minimum number of available RBs is enabled, dynamic RB control cannot be enabled, that is, the RB_DYNAMIC_CONTROL_SW option of the NRDUCellAlgoSwitch.HighReliability-Switch parameter cannot be selected.
Low-frequency TDD	QCI-based resource control	RB_DYNAMIC_CONTROL_SW option of the NRDUCellFeatureSw.QciBasedDiffServExpSw parameter	<i>QCI-based Resource Control</i>	When QCI-based resource control is enabled, dynamic RB control cannot be enabled, that is, the RB_DYNAMIC_CONTROL_SW option of the NRDUCellAlgoSwitch.HighReliability-Switch parameter cannot be selected.
Low-frequency TDD	FSD transmission mode	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL NRDUCellMultiTrp.DataTransMode set to FSD_MODE	<i>Hyper Cell</i>	In FSD transmission mode, dynamic RB control (controlled by the RB_DYNAMIC_CONTROL_SW option of the NRDUCellAlgoSwitch.HighReliability-Switch parameter) cannot be enabled.

6.1.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink interference randomization-based scheduling	DL_INTRF_RANDOM_SWITCH option of the NRDUCellAlgoSwitch.AdaptiveEdgeEnhanceSwitch parameter	<i>Downlink Scheduling</i>	In independent resource reservation in non-slicing scenarios, downlink interference randomization-based scheduling does not take effect if NRDUCellRes.DlMinRbStartPos is set to 0.
Low-frequency TDD	Uplink interference randomization-based scheduling	NRDUCellAlgoSwitch.UlInterfRandomSwitch	<i>Uplink Scheduling</i>	In independent resource reservation in non-slicing scenarios, uplink interference randomization-based scheduling does not take effect if NRDUCellRes.UlMinRbStartPos is set to 0.
Low-frequency TDD	Interference avoidance for indoor cells	INDOOR_INTERF_AVOID_SWITCH option of the NRDUCellAlgoSwitch.IndoorOutdoorCooperateSwitch parameter	<i>3D Networking Experience Improvement</i>	In independent resource reservation in non-slicing scenarios, interference avoidance for indoor cells does not take effect if NRDUCellRes.DlMinRbStartPos is set to 0.
Low-frequency TDD	Enhanced interference avoidance for indoor cells	INDOOR_INTERF_AVOID_ENH_SWITCH option of the NRDUCellAlgoSwitch.IndoorOutdoorCooperateSwitch parameter	<i>3D Networking Experience Improvement</i>	When NRDUCellRes.DlMinRbStartPos is set to 0, enhanced interference avoidance for indoor cells does not take effect.
Low-frequency TDD	SRS Po adaptation	PO_SRS_ADAPT_SWITCH option of the NRDUCellULPcConfig.UlPwrCtrlAlgoSwitch parameter	<i>Power Control</i>	When NRDUCellSrs.UllcPoOffset is set to a value other than 0 for uplink differentiated power control, SRS Po adaptation does not take effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink resource isolation	DL_EMBB_PREFERENTIAL_SCH_SW option of the NRDUCellAlgoSwitch.FwaAlgoSwitch parameter	<i>FWA</i>	After downlink resource isolation is enabled: If the minimum proportion of downlink RBs in dynamic RB control for UEs requiring low latency and high reliability has taken effect for FWA users, downlink resource isolation will not take effect for these FWA users; If the minimum proportion of downlink RBs in dynamic RB control for UEs requiring low latency and high reliability has taken effect for non-FWA users, the average downlink throughput of FWA users decreases more severely.
Low-frequency TDD	Downlink multi-BWP interference avoidance	DL_BWP_HYBRID_INTRF_RANDOM_SW option of the NRDUCellDisch.DISchAlgoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	When power saving BWP is enabled together with the function of low latency and high reliability, UEs requiring connectivity with low latency and high reliability will not switch to BWP2. In addition, as the function of low latency and high reliability has a higher priority, RB reservation will take effect, thereby causing the benefits of downlink multi-BWP interference avoidance to decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink SU-MIMO	UL_SU_MULTILAYER_SW option of the NRDUCellAlgoSwitch.SuMIMOMultipleLayerSw parameter	<i>MIMO (TDD)</i>	The base station can adjust the maximum rank for UEs requiring connectivity with low latency and high reliability based on the value of the gNBDUSchParamGroup.UrlcMaxRank parameter.
Low-frequency TDD	Downlink SU-MIMO	DL_SU_MULTILAYER_SW option of the NRDUCellAlgoSwitch.SuMIMOMultipleLayerSw parameter		
Low-frequency TDD	Coordinated interference management	INTRA_GNB_DL_CS_CBF_SW and INTER_GNB_DL_CS_CBF_SW options of the NRDUCellAlgoSwitch.CoopSwitch parameter	<i>Coordinated Interference Management (Low-Frequency TDD)</i>	RB0 may not be the start RB for scheduling UEs within the minimum RB quota in a cooperating cell when both of the following conditions are met: (1) Coordinated interference management is enabled; (2) The start RB of the minimum RB quota (for resource reservation) is configured. (Currently, only RB0 can be configured as the start RB.)
Low-frequency TDD	Full-bandwidth initial BWP configuration	INIT_BW_FULL_BW_SW option of the NRDUCellAlgoSwitch.BwpConfigPolicySwitch parameter	<i>5G Networking and Signaling</i>	When uplink interference randomization-based scheduling within fixed resources is enabled, a full-bandwidth initial BWP is recommended.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	External interference avoidance	PUCCH_INTRF_AVOID_SW option of the NRDUCellAlgoSwitch.CommChnlIntrfAvoidSwitch parameter	<i>Interference Avoidance</i>	When the range experiencing external interference conflicts with the fixed resources reserved for services requiring connectivity with high reliability, you are advised not to configure fixed resource reservation for such services.
Low-frequency TDD	NR DU cell resource management between network slices	NRDUCellAlgoSwitch.NetworkSliceAlgoSwitch	<i>Network Slicing</i>	Only a single UE is supported in a scenario where all the following conditions are met: (1) Both NR DU cell resource management between network slices and QCI-based uplink precise preallocation are enabled; (2) NRDUCellRes.UlRbMaxRatio is set to a value indicating 1%; (3) NRDUCellRes.ResStrategyType is set to TTI_ONLY . This is because, in this scenario, reserved resources are preferentially used per slot for the first UE that is running a service requiring connectivity with high reliability, which will lead to failed scheduling of reconfiguration completion messages for other UEs in the slice group.

6.1.3.3 Hardware

For details, see [4.3.3 Hardware](#).

6.1.3.4 Cells

For details, see [4.3.4 Cells](#).

6.1.3.5 Others

UE support required in data channel reliability optimization:

- MCS index table 3: requires UEs to support MCS index tables that incorporate low code rates. This support is indicated by the **dl-64QAM-MCS-TableAlt** and **ul-64QAM-MCS-TableAlt** fields of the *Phy-ParametersFRX-Diff* IE in the RRC_UE_CAP_INFO message. You can verify UE support using this RRC message. (This message is sent over the air interface and can be observed during a network access procedure.)
- CQI table 3: requires UE support. A UE indicates whether it supports CQI table 3 only after receiving both CQI table 3 and MCS index table 3 from the gNodeB. This support is indicated by the **cqi-TableAlt** field of the *Phy-ParametersFRX-Diff* IE in the RRC_UE_CAP_INFO message. You can verify UE support using this RRC message. (This message is sent over the air interface and can be observed during a network access procedure.)
- PDCP out-of-order delivery: requires UE support. This support is indicated by the **outOfOrderDelivery** field of the *PDCP-Parameters* IE in the RRC_UE_CAP_INFO message. You can verify UE support using this RRC message. (This message is sent over the air interface and can be observed during a network access procedure.)
- Other functions: none

6.1.4 Operation and Maintenance

6.1.4.1 Data Configuration

6.1.4.1.1 Data Preparation

[Table 6-5](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 6-5 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
High Reliability Switch	NRDUCellAlgoSwitch.HighReliabilitySwitch	HIGH_RELIABILITY_BASIC_SW	To enable low latency and high reliability, select this option.

Parameter Name	Parameter ID	Option	Setting Notes
High Reliability Switch	NRDUCellAlgoSwitch.HighReliabilitySwitch	RB_DYNAMIC_CONTROL_SW	To enable resource reservation for low latency and high reliability, select this option.
Service Differentiation Algorithm Switch	NRDUCellQciBearer.ServiceDiffAlgorithmSwitch	CQI_TABLE3_MEAS_SW	To use CQI table 3, select this option.
Service Differentiation Algorithm Switch	NRDUCellQciBearer.ServiceDiffAlgorithmSwitch	MCS_TABLE3_SW	To use MCS index table 3, select this option.
Scheduling Type	gNBDUMacParamGroup.SchedulingType	None	To enable high-reliability-oriented scheduling, set this parameter to HIGH_RELIABILITY_SCH .
Service Type	gNBQciBearer.ServiceType	None	To enable high-reliability-oriented scheduling, set this parameter to URLLC_HIGH_RELIABILITY_SERVICE .
Resource Usage Object	NRDUCellRes.ResourceUsageObject	None	To apply a resource template to high reliability, set this parameter to HIGH_RELIABILITY .
Cell Resource ID	NRDUCellQciBearer.CellResId	None	Set this parameter to the cell resource ID for a resource template.
Resource Strategy Type	NRDUCellRes.ResourceStrategyType	None	Set this parameter to TTI_ONLY .
Uplink Min RB Start Position	NRDUCellRes.UplinkMinRbStartPos	None	Set this parameter to 0 .
Uplink RB Max Ratio	NRDUCellRes.UplinkRbMaxRatio	None	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
Uplink RB Min Ratio	NRDUCellRes.UlRbMinRatio	None	Set this parameter based on the network plan.
Downlink Min RB Start Position	NRDUCellRes.DlMinRbStartPos	None	Set this parameter to 0 .
Downlink RB Max Ratio	NRDUCellRes.DlRbMaxRatio	None	Set this parameter based on the network plan.
Downlink RB Min Ratio	NRDUCellRes.DlRbMinRatio	None	Set this parameter based on the network plan.
BWP Configuration Policy Switch	NRDUCellAlgoSwitch.BwpConfigPolicySwitch	INIT_BWP_FULL_BW_SW	Select this option.
gNodeB PDCP Out Of Order Delivery Ind	gNBpdcpparamgroup.GnbPdcppOutOfOrderDeliveryInd	None	Set this parameter to TRUE_WITHOUT_UE_CAPB if the gNodeB is required to support out-of-order PDCP packet delivery to the upper layer.
UE PDCP Out Of Order Delivery Ind	gNBpdcpparamgroup.UePdcppOutOfOrderDeliveryInd	None	Set this parameter to TRUE if a UE is required to support out-of-order PDCP packet delivery to the upper layer.
RLC Mode	NRCCellQciBearer.RlcMode	None	Set this parameter based on the network plan.
Uplink Maximum HARQ Retrans Number	gNBDUSchParamGroup.UlMaxHarqRetransNum	None	Set this parameter based on the network plan.
Downlink Maximum HARQ Retrans Number	gNBDUSchParamGroup.DlMaxHarqRetransNum	None	Set this parameter based on the network plan.
Uplink Initial MCS Adjust Value	gNBDUSchParamGroup.UlInitialMcsAdjValue	None	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
Downlink Initial MCS Adjust Value	gNBDUSchParamGroup.DlInitialMcsAdjValue	None	Set this parameter based on the network plan.
Uplink Maximum MCS	gNBDUSchParamGroup.UlMaxMcs	None	Set this parameter based on the network plan.
Uplink Minimum MCS	gNBDUSchParamGroup.UlMinMcs	None	Set this parameter based on the network plan.
Uplink AMC Adjustment Step	gNBDUSchParamGroup.UlAmcAdjustStep	None	Set this parameter based on the network plan.
Downlink AMC Adjustment Step	gNBDUSchParamGroup.DlAmcAdjustStep	None	Set this parameter based on the network plan.
Downlink Maximum MCS	gNBDUSchParamGroup.DlMaxMcs	None	Set this parameter based on the network plan.
Downlink Minimum MCS	gNBDUSchParamGroup.DlMinMcs	None	Set this parameter based on the network plan.
Low Latency Switch	NRDUCellAlgoSwitch.LowLatencySwitch	SERV_DIFF_LOW_LATENCY_SW	To enable service-differentiated low latency, select this option.
Schedule Option Switch	gNBDUSchParamGroup.ScheduleOptionSwitch	UL_PREALLOCATION_QCI_SW	To enable QCI-based uplink preallocation, select this option.
Schedule Option Switch	gNBDUSchParamGroup.ScheduleOptionSwitch	UL_IBLER_QCI_SW	To enable QCI-specific scheduling based on the uplink target IBLER, select this option.
Schedule Option Switch	gNBDUSchParamGroup.ScheduleOptionSwitch	DL_IBLER_QCI_SW	To enable QCI-specific scheduling based on the downlink target IBLER, select this option.

Parameter Name	Parameter ID	Option	Setting Notes
Uplink Preallocation Interval	gNBDUSchParamGroup.UlPreallocationInterval	None	After QCI-based uplink preallocation is enabled, set this parameter based on the network plan.
Uplink Preallocation Data Volume	gNBDUSchParamGroup.UlPreallocationDataVolume	None	After QCI-based uplink preallocation is enabled, set this parameter based on the network plan. It is recommended that this parameter be set to twice the service packet size.
Uplink Target IBLER	gNBDUSchParamGroup.UlTargetIbler	None	Set this parameter based on the network plan.
Downlink Target IBLER	gNBDUSchParamGroup.DlTargetIbler	None	Set this parameter based on the network plan.
RLC Mode	gNBRlcParamGroup.RlcMode	None	Set this parameter based on the network plan.
Radio Bearer Type	gNBRlcParamGroup.RadioBearerType	None	Set this parameter based on the network plan.
Polling PDU Retransmission Timer for gNodeB	gNBRlcParamGroup.gNBPollingPduRetransTimer	None	Set this parameter based on the network plan.
Polling PDU Retransmission Timer for UE	gNBRlcParamGroup.UePollingPduRetransTimer	None	Set this parameter based on the network plan.
RLC Reassembly Timer for gNodeB	gNBRlcParamGroup.gNBRlcReassemblyTimer	None	Set this parameter based on the network plan.
RLC Reassembly Timer for UE	gNBRlcParamGroup.UeRlcReassemblyTimer	None	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
AM Status Report Prohibit Timer for gNodeB	gNBRLcParamGroup.gNBAmStatusReportProhibitTmr	None	Set this parameter based on the network plan.
AM Status Report Prohibit Timer for UE	gNBRLcParamGroup.UeAmStatusReportProhibitTmr	None	Set this parameter based on the network plan.
Scheduling Policy Switch	NRDUCellRes.SchedulingPolicySwitch	FIX_RES_UL_IN TRF_RANDOM_SCH_SW	To enable uplink interference randomization-based scheduling within fixed resources, select this option.
URLLC PO Offset	NRDUCellSrs.UrllicPoOffset	None	Set this parameter based on the network plan.
URLLC Max Rank	gNBDUSchParamGroup.UrllicMaxRank	None	It is recommended that this parameter be set to RANK2 when frequency-domain resources are insufficient and to RANK1 in other scenarios.
MLB Trigger Mode	NRCeIlMlb.MlbTriggerMode	None	Set this parameter to CELL_USER_NUM in load balancing scenarios.
QCI Sw Based UE Selection Policy Pri	NRCeIlMlb.QciSwBasedUeSelPolicyPri	None	Set this parameter to 255 in load balancing scenarios.
MLB Algorithm Switch	NRCeIlQciBearer.MlbAlgoSwitch	MLB_FILTER_SW	Select this option in load balancing scenarios.
Uplink PUSCH Algorithm Switch	NRDUCellPusch.UlPuschAlgoSwitch	PUSCH_ACK_DTX_DETECT_SW	Select this option.

6.1.4.1.2 Using MML Commands

Before using MML commands, refer to [6.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual

network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

NOTICE

If (1) resource reservation is used to guarantee the availability of data channel resources for services requiring connectivity with high reliability and (2) resources are allocated on a per RBG basis, the maximum number of available RBs must be greater than or equal to the number of RBs in one RBG for both UEs requiring connectivity with high reliability and other UEs. To ensure that this requirement is met, all of the following conditions must be met:

- Number of RBs corresponding to **NRDUCellRes.DIRbMaxRatio** or **NRDUCellRes.UIRbMaxRatio** $\geq$ Number of RBs in an RBG
- Total number of available PDSCH RBs in a cell – Number of RBs corresponding to **NRDUCellRes.DIRbMinRatio** for UEs requiring connectivity with high reliability $\geq$ Number of RBs in an RBG
- Total number of available PUSCH RBs in a cell – Number of RBs corresponding to **NRDUCellRes.UIRbMinRatio** for UEs requiring connectivity with high reliability $\geq$ Number of RBs in an RBG

Activation Command Examples

```
//Turning on the switch for basic high-reliability functions
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, HighReliabilitySwitch=HIGH_RELIABILITY_BASIC_SW-1;

//Adding a high-reliability-oriented bearer (Configuration with a single bearer of 5QI 82 requires support
from the core network.)
ADD GNB5QICONFIG: Nr5qi=82, Qci=82;
ADD GNBQCIBEARER: Qci=82;
//Setting gNBQciBearer.ServiceType to URLLC_HIGH_RELIABILITY_SERVICE for a QCI of a high-reliability-
oriented bearer on the base station (Otherwise, low latency and high reliability will not take effect for the
QCI-specific bearer.)
MOD GNBQCIBEARER: Qci=82, ServiceType=URLLC_HIGH_RELIABILITY_SERVICE;

//Configuring a service requiring connectivity with high reliability, with a high-reliability-oriented bearer
configured and associated with a QCI
ADD NRCELLQCIBEARER: NrCellId=1, Qci=82;
ADD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=82;
MOD GNBDMACPARAMGROUP: MacParamGroupId=1, SchedulingType=HIGH_RELIABILITY_SCH;
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=82, MacParamGroupId=1;

//Turning on the switch for CQI table 3-based measurement and the switch for MCS index table 3
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=82,
ServiceDiffAlgoSwitch=CQI_TABLE3_MEAS_SW-1&MCS_TABLE3_SW-1;

//Disabling the DRX function for a specific QCI
MOD NRCELLQCIBEARER: NrCellId=1, Qci=82, DrxParamGroupId=255;
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=82, DuDrxParamGroupId=255;

//Configuring the QCI Sw Based UE Selection Policy Pri parameter in the event that load balancing is used
MOD NRCELLMLB: NrCellId=1, MlbTriggerMode=CELL_USER_NUM, QciSwBasedUeSelPolicyPri=255;
//Selecting the MLB_FILTER_SW option for the QCI specific to services without MLB involved
MOD NRCELLQCIBEARER: NrCellId=1, Qci=82, MlbAlgoSwitch=MLB_FILTER_SW-1;
```

```
//Selecting the PUSCH_ACK_DTX_DETECT_SW option  
MOD NRDUCELLPUSCH: NrDuCellId=1, UIPuschAlgoSwitch=PUSCH_ACK_DTX_DETECT_SW-1;
```

For details about configuration of slicing-based resource reservation, see *Network Slicing*.

Independent resource reservation in non-slicing scenarios

```
//Adding a high-reliability bearer and a resource template and associating the high-reliability bearer with  
the resource template  
ADD NRDUCELLRES: NrDuCellId=1, CellResId=1, ResUsageObject=HIGH_RELIABILITY,  
ResStrategyType=TTI_ONLY, UIMinRbStartPos=0, DMinRbStartPos=0, UIRbMinRatio=10, UIRbMaxRatio=30,  
DIRbMinRatio=10, DIRbMaxRatio=30;  
ADD NRCELLQCIBEARER: NrCellId=1, Qci=82;  
ADD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=82;  
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=82, CellResId=1;  
  
//Enabling resource reservation for high-reliability services in non-slicing scenarios  
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, HighReliabilitySwitch=RB_DYNAMIC_CONTROL_SW-1;  
  
//(Optional) Configuring fixed resource reservation (recommended when a neighboring cell causes  
interference to a cell where high-reliability services are running)  
//Configuration for the cell where high-reliability services are running  
MOD NRDUCELLRES: NrDuCellId=1, CellResId=1, ResUsageObject=HIGH_RELIABILITY,  
ResStrategyType=TTI_ONLY, UIMinRbStartPos=0, DMinRbStartPos=0, UIRbMinRatio=10, UIRbMaxRatio=30,  
DIRbMinRatio=10, DIRbMaxRatio=30;  
//Configuration for the interfering neighboring cell  
ADD NRDUCELLRES: NrDuCellId=2, CellResId=1, ResUsageObject=HIGH_RELIABILITY, DIRbMaxRatio=30,  
DIRbMinRatio=10, UIRbMaxRatio=30, UIRbMinRatio=10, DMinRbStartPos=0, UIMinRbStartPos=0;  
ADD NRCELLQCIBEARER: NrCellId=2, Qci=82;  
ADD NRDUCELLQCIBEARER: NrDuCellId=2, Qci=82;  
MOD NRDUCELLALGOSWITCH: NrDuCellId=2, BwpConfigPolicySwitch=INIT_BWP_FULL_BW_SW-1;  
MOD NRDUCELLQCIBEARER: NrDuCellId=2, Qci=82, CellResId=2;  
  
//(Optional) Configuring fixed resource reservation and enabling uplink interference randomization-based  
scheduling within fixed resources (recommended when neighboring cells cause interference to multiple cells  
where high-reliability services are running)  
//Configuration for the cell where high-reliability services are running (NrDuCellId=1 is used as an example)  
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, HighReliabilitySwitch=RB_DYNAMIC_CONTROL_SW-1;  
MOD NRDUCELLRES: NrDuCellId=1, CellResId=1, ResUsageObject=HIGH_RELIABILITY,  
ResStrategyType=TTI_ONLY, UIMinRbStartPos=0, DMinRbStartPos=0, UIRbMinRatio=30, UIRbMaxRatio=30,  
DIRbMinRatio=30, DIRbMaxRatio=30, SchedulingPolicySwitch=FIX_RES_UL_INTRF_RANDOM_SCH_SW-1;  
//Configuration for the interfering neighboring cell  
ADD NRDUCELLRES: NrDuCellId=2, CellResId=1, ResUsageObject=HIGH_RELIABILITY, DIRbMaxRatio=60,  
DIRbMinRatio=30, UIRbMaxRatio=60, UIRbMinRatio=30, DMinRbStartPos=0, UIMinRbStartPos=0;  
MOD NRDUCELLQCIBEARER: NrDuCellId=2, Qci=82, CellResId=2;
```

OLLA algorithm optimization

```
//Enabling OLLA algorithm optimization  
ADD GNBDSCHPARAMGROUP: ScheduleParamGroupId=1;  
MOD GNBDSCHPARAMGROUP: ScheduleParamGroupId=1, UInitialMcsAdjValue=-4,  
DInitialMcsAdjValue=-4, UIMaxHarqRetransNum=4, DMaxHarqRetransNum=4, UIMaxAdjustStep=10,  
DMaxAdjustStep=10, UIMaxMcs=23, DMaxMcs=23, UIMinMcs=0, DMinMcs=0;  
//Mapping the gNodeB DU scheduling parameter group to the bearer of a specific QCI  
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=82, ScheduleParamGroupId=1;
```

PDCP out-of-order delivery

```
//Enabling PDCP out-of-order delivery  
MOD NRCELLQCIBEARER: NrCellId=1, Qci=82, AmPdcpparamGroupId=9;  
MOD GNBPDCCPARAMGROUP: PdcpparamGroupId=9,  
GnbPdcppOutOfOrderDeliveryInd=TRUE_WITHOUT_UE_CAPB, UePdcppOutOfOrderDeliveryInd=TRUE;
```

Service-differentiated low latency

```
//Enabling service-differentiated low latency  
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, LowLatencySwitch=SERV_DIFF_LOW_LATENCY_SW-1;  
//Adding a gNodeB DU scheduling parameter group
```

```
ADD GNBUSCHPARAMGROUP: ScheduleParamGroupId=1;
//Enabling QCI-based uplink preallocation and configuring the interval and data volume for preallocation
MOD GNBUSCHPARAMGROUP: ScheduleParamGroupId=1,
ScheduleOptSwitch=UL_PREALLOCATION_QCI_SW-1, UIPreallocationInterval=1,
UIPreallocationDataVolume=100;
//Setting the uplink and downlink target IBLERs (1% is recommended)
MOD GNBUSCHPARAMGROUP: ScheduleParamGroupId=1,
ScheduleOptSwitch=UL_IBLER_QCI_SW-1&DL_IBLER_QCI_SW-1, UITargetIbler=1000, DITargetIbler=1000;
//Mapping the gNodeB DU scheduling parameter group to the bearer of a specific QCI
MOD NRDUCELLQCI BEARER: NrDuCellId=1, Qci=82, ScheduleParamGroupId=1;
```

HARQ retransmission optimization

```
//Enabling HARQ retransmission optimization and selecting the RLC transmission mode based on service
requirements
//Method 1: Selecting the AM mode
ADD GNBRLCPARAMGROUP: RlcParamGroupId=82, RlcMode=AM, RadioBearerType=DRB;
MOD NRDUCELLQCI BEARER: NrDuCellId=1, Qci=82, AmRlcParamGroupId=82;
MOD GNBUSCHPARAMGROUP: ScheduleParamGroupId=1, UIMaxHarqRetransNum=2,
DlMaxHarqRetransNum=2;
MOD GNBRLCPARAMGROUP: RlcParamGroupId=82, RlcMode=AM, RadioBearerType=DRB,
UePollingPduRetransTimer=MS15, gNBPollingPduRetransTimer=MS15, UeAmStatusRptProhibitTmr=MS15,
gNBAmStatusRptProhibitTmr=MS15, gNBRLCReassemblyTimer=MS15, UeRlcReassemblyTimer=MS15;
//Method 2: Selecting the UM mode
ADD GNBRLCPARAMGROUP: RlcParamGroupId=82, RlcMode=UM, RadioBearerType=DRB;
MOD NRDUCELLQCI BEARER: NrDuCellId=1, Qci=82, UmRlcParamGroupId=82;
MOD GNBUSCHPARAMGROUP: ScheduleParamGroupId=1, UIMaxHarqRetransNum=2,
DlMaxHarqRetransNum=2;
```

Uplink differentiated power control

```
//Configuring a PO offset for UEs requiring connectivity with low latency and high reliability
MOD NRDUCELLSRS: NrDuCellId=1, UrllcPoOffset=0;
```

Rank configuration for low latency and high reliability

```
//(Optional) Setting the maximum rank for URLLC (RANK2 is recommended when frequency-domain
resources are insufficient and RANK1 is recommended in other scenarios)
MOD GNBUSCHPARAMGROUP: ScheduleParamGroupId=1, UrllcMaxRank=RANK1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

Independent resource reservation in non-slicing scenarios

```
//Disabling resource reservation for high-reliability services in non-slicing scenarios
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, HighReliabilitySwitch=RB_DYNAMIC_CONTROL_SW-0;
//(Optional) Turning off the switch for uplink interference randomization-based scheduling
MOD NRDUCELLRES: NrDuCellId=1, CellResId=1, ResUsageObject=HIGH_RELIABILITY,
SchedulingPolicySwitch=FIX_RES_UL_INTRF_RANDOM_SCH_SW-0;
```

PDCP out-of-order delivery

```
//Disabling PDCP out-of-order delivery
MOD GNBPDCCPARAMGROUP: PdcpparamGroupId=9, GnbPdcppOutOfOrderDeliveryInd=FALSE,
UePdcppOutOfOrderDeliveryInd=FALSE;
```

Service-differentiated low latency

```
//Disabling service-differentiated low latency
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, LowLatencySwitch=SERV_DIFF_LOW_LATENCY_SW-0;
```

Rank configuration for low latency and high reliability

```
//(Optional) Setting the maximum rank used in the uplink and downlink to NOT_CONFIG for UEs requiring
connectivity with low latency and high reliability
MOD GNBUSCHPARAMGROUP: ScheduleParamGroupId=1, UrllcMaxRank=NOT_CONFIG;
```

Switch for basic high-reliability functions

```
//Turning off the switch for basic high-reliability functions  
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, HighReliabilitySwitch=HIGH_RELIABILITY_BASIC_SW-0;
```

6.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.1.4.2 Activation Verification

To check whether functions related to data channel reliability optimization have been activated, perform real-time UE-level performance monitoring as follows:

- Step 1** After a UE accesses a cell, enable the UE to set up a high-reliability-oriented bearer. Move the UE to the cell center and enable the UE to perform a downlink ping service instead of an uplink service.
- Step 2** On the MAE-Access, start UE-level channel quality monitoring (corresponding to **Quality of Channel Monitoring** on the GUI) and MCS-based count monitoring (corresponding to **MCS Count Monitoring** on the GUI).
- Step 3** Observe the UE channel quality by **Quality of Channel Monitoring** and the number of uplink scheduling times by **MCS Count Monitoring**. If the number of uplink scheduling times increases, functions related to data channel reliability optimization have taken effect.

----End

6.1.4.3 Network Monitoring

For details, see [4.4.3 Network Monitoring](#).

6.2 Control Channel Reliability Optimization

6.2.1 Principles

Control channel reliability optimization includes target PDCCH IBLER optimization, PDCCH CCE optimization, and PDCCH aggregation level adjustment mechanism optimization. The objective of these optimizations is to enhance the reliability of the PDCCH.

Target PDCCH IBLER Optimization

A target IBLER for PDCCH link adaptation can be specifically configured for each QCI of UEs requiring connectivity with low latency and high reliability. A reduction in the target IBLER results in a decrease in the PDCCH BLER, thereby enhancing PDCCH reliability for UEs requiring connectivity with low latency and high reliability.

You can configure the **gNBDUSchParamGroup.PdcchBlerTarget** parameter to specify a target IBLER.

- A smaller value of this parameter results in a lower PDCCH BLER.
- A larger value of this parameter results in a higher PDCCH BLER.

This function takes effect only for VoNR UEs (UEs with 5QI-1 bearers) and UEs requiring connectivity with high reliability (UEs with **gNBQciBearer.ServiceType** set to **URLLC_HIGH_RELIABILITY_SERVICE**).

NOTE

If a QCI is not defined in the **NRDUCellQciBearer** MO, the target IBLER for the QCI is determined by the **NRDUCellPdcch.PdcchBlerTarget** parameter rather than the **gNBDUSchParamGroup.PdcchBlerTarget** parameter associated with the **NRDUCellQciBearer.Qci** parameter that has a value of 9.

PDCCH CCE Optimization

When there are a large number of online UEs, control channel elements (CCEs) may be insufficient for services requiring connectivity with high reliability. As a result, a wait period is required before the scheduling of UEs running such services can be performed, which increases delay. In such scenarios, it is recommended that PDCCH CCE optimization be enabled. This involves the following operations:

- Setting the **NRDUCellPdcch.OccupiedSymbolNum** parameter to **2SYM** to double CCE resources. This parameter indicates the number of symbols occupied by the PDCCH.
- Enabling PDCCH uplink-to-downlink CCE ratio adaptation (by selecting the **UL_DL_CCE_RATIO_ADAPT_SW** option of the **NRDUCellPdcch.PdcchAlgoSwitch** parameter). This enables the ratio of available CCEs for uplink scheduling information to those for downlink scheduling information to be adaptively adjusted based on specific factors, including the CCEs required for uplink and downlink scheduling information and CCE usage. For details about PDCCH uplink-to-downlink CCE ratio adaptation, see *Channel Management*.
- Enabling SR-based smart scheduling (by selecting the **SR_SMART_SCH_SW** option of the **NRDUCellPusch.UlPuschAlgoSwitch** parameter). This enables the gNodeB to prioritize the SRs of the UEs that are at risk of service drops, followed by the UEs that are accessing the network and then other UEs, in terms of SR scheduling.

PDCCH Aggregation Level Adjustment Mechanism Optimization

PDCCH aggregation level adjustment mechanism optimization is introduced to improve control channel reliability for UEs requiring connectivity with high reliability. **This function allows you to modify the minimum PDCCH aggregation level by using the gNBDUSchParamGroup.PdcchMinAggLvl parameter. The minimum PDCCH aggregation level can be increased to a proper level. This increases blind detections with higher PDCCH aggregation levels while restricting the number of blind detections to the upper limit. Consequently, the probability of missed PDCCH detections is reduced.**

This function takes effect only for VoNR UEs (UEs with 5QI-1 bearers) and UEs requiring connectivity with high reliability (UEs with **gNBQciBearer.ServiceType** set to **URLLC_HIGH_RELIABILITY_SERVICE**).

It is recommended that this function be used only in the following scenarios and configurations:

- Special WAN: On a WAN with a small number of UEs requiring connectivity with high reliability, it is recommended that the minimum PDCCH aggregation level be set to 4 for these UEs. An example of such a WAN is a power grid network for distribution automation where there are typically one to four such UEs in a cell, with the number of other UEs being less than 80.
- ToB private network: On this kind of network, if 12 or fewer UEs require connectivity with high reliability, it is recommended that **NRDUCellPdcch.OccupiedSymbolNum** be set to **2SYM** or **3SYM** and the minimum PDCCH aggregation level be set to 4 for these UEs.

6.2.2 Network Analysis

6.2.2.1 Benefits

The following table lists the benefits of control channel reliability optimization.

Function	Benefit
Target PDCCH BLER optimization	This function increases the PDCCH SINR to a certain level, thereby decreasing the BLER.
PDCCH CCE optimization	This function increases the number of PDCCH CCEs available to services requiring connectivity with low latency and high reliability, thereby reducing the proportion of type-1 SRs.
PDCCH aggregation level adjustment mechanism optimization	This function reduces the probability of missed PDCCH detection, thereby shortening delay and improving reliability.

Conditions for Optimal Gains

For details, see [Conditions for Optimal Gains](#).

6.2.2.2 Impacts

The impacts between this function and other functions are described in [6.2.3.2.3 Function Impacts](#).

Functions related to control channel reliability optimization cause a high consumption of uplink and downlink resources due to high-reliability-oriented scheduling being applied to UEs requiring connectivity with low latency and high reliability. This reduces the resources available to other UEs.

Function	Impact on the Network
Target PDCCH IBLER optimization	No impact
PDCCH CCE optimization	Changing the value of the NRDUCellPdcch.OccupiedSymbolNum parameter causes the cell to restart and all UEs served by the cell to re-access the network.
PDCCH aggregation level adjustment mechanism optimization	No impact

6.2.3 Requirements

6.2.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-051206	URLLC Introduction	NR0SURLLCP00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

6.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.2.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Identification of UEs requiring connectivity with low latency and high reliability	HIGH_RELIABILITY_BASIC_SWITCH.HighReliabilitySwitch parameter gNBDUMacParamGroup.SchedulingType set to HIGH_RELIABILITY_SCH gNBQciBearer.ServiceType set to URLLC_HIGH_RELIABILITY_SERVICE	<i>URLLC</i>	The following sub-functions of control channel reliability optimization depend on identification of UEs requiring connectivity with low latency and high reliability: • Target PDCCH IBLER optimization • PDCCH aggregation level adjustment mechanism optimization

6.2.3.2.2 Mutually Exclusive Functions

None

6.2.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Maximum PDCCH CCE aggregation level	NRDUCellPdcchAlgo.PdcchMaxAggLevel set to AGG_LVL_4	<i>Channel Management</i>	For UEs for which the configured minimum PDCCH aggregation level takes effect, if the configured maximum PDCCH CCE aggregation level is lower than the configured minimum aggregation level, the maximum PDCCH CCE aggregation level takes effect instead of the configured value. That is, for these UEs, if the NRDUCellPdcchAlgo.PdcchMaxAggLevel parameter is set to AGG_LVL_4 and the gNBDUSchParamGroup.PdcchMinAggLvl parameter is set to AGG_LVL_8 , the value AGG_LVL_16 of the NRDUCellPdcchAlgo.PdcchMaxAggLevel parameter takes effect.

6.2.3.3 Hardware

For details, see [4.3.3 Hardware](#).

6.2.3.4 Cells

For details, see [4.3.4 Cells](#).

6.2.3.5 Others

None

6.2.4 Operation and Maintenance

6.2.4.1 Data Configuration

6.2.4.1.1 Data Preparation

Table 6-6 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 6-6 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
High Reliability Switch	NRDUCellAlgoSwitch.HighReliabilitySwitch	HIGH_RELIABILITY_BASIC_SW	Select this option.
Scheduling Type	gNBDUMacParamGroup.SchedulingType	None	To enable high-reliability-oriented scheduling, set this parameter to HIGH_RELIABILITY_SCH .
Service Type	gNBQciBearer.ServiceType	None	To enable high-reliability-oriented scheduling, set this parameter to URLLC_HIGH_RELIABILITY_SERVICE .
PDCCH BLER Target	gNBDUSchParamGroup.PdcchBlerTarget	None	Set this parameter based on the network plan.
Occupied Symbol Number	NRDUCellPdcch.OccupiedSymbolNumber	None	Set this parameter based on the network plan.
PDCCH Algorithm Switch	NRDUCellPdcch.PdcchAlgoSwitch	UL_DL_CCE_RATIO_ADAPT_SW	To enable PDCCH CCE optimization, select this option.
Uplink PUSCH Algorithm Switch	NRDUCellPusch.UlPuschAlgoSwitch	SR_SMART_SCH_SW	To enable SR-based smart scheduling, select this option.

Parameter Name	Parameter ID	Option	Setting Notes
PDCCH Min Aggregation Level	gNBDUSchParamGroup.PdcchMinAggLvl	None	It is recommended that PDCCH aggregation level adjustment mechanism optimization be implemented only in certain scenarios. The following are recommended scenarios and configurations for UEs requiring connectivity with high reliability: • Special WAN: On a WAN with a small number of UEs requiring connectivity with high reliability, it is recommended that the minimum PDCCH aggregation level be set to 4 for these UEs. One such example is a power grid network for distribution automation, where there are typically one to four such UEs in a cell, with the number of ToC background UEs less than 80. • ToB private network: On this kind of network, if 12 or fewer UEs require connectivity with high reliability, it is recommended that NRDUCellPdcch.OccupiedSymbolNum be set to 2SYM

Parameter Name	Parameter ID	Option	Setting Notes
			or 3SYM and the minimum PDCCH aggregation level be set to 4 for these UEs. If the maximum aggregation level configured in the <i>ControlResourceSet</i> IE (see 3GPP TS 38.331) for a UE is less than the value of this parameter, the actual aggregation level used for the UE will be less than the value of this parameter.

6.2.4.1.2 Using MML Commands

Before using MML commands, refer to [6.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the switch for basic high-reliability functions
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, HighReliabilitySwitch=HIGH_RELIABILITY_BASIC_SW-1;

//Adding a high-reliability-oriented bearer (Configuration with a single bearer of 5QI 82 requires support
from the core network.)
ADD GNB5QICONFIG: Nr5qi=82, Qci=82;
ADD GNBQCIBEARER: Qci=82;
//Setting gNBQciBearer.ServiceType to URLLC_HIGH_RELIABILITY_SERVICE for a QCI of a high-reliability-
oriented bearer on the base station (Otherwise, low latency and high reliability will not take effect for the
QCI-specific bearer.)
MOD GNBQCIBEARER: Qci=82, ServiceType=URLLC_HIGH_RELIABILITY_SERVICE;

//Configuring a service requiring connectivity with high reliability, with a high-reliability-oriented bearer
configured and associated with a QCI
ADD NRCELLQCIBEARER: NrCellId=1, Qci=82;
ADD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=82;
MOD GNBUMACPARAMGROUP: MacParamGroupId=1, SchedulingType=HIGH_RELIABILITY_SCH;
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=82, MacParamGroupId=1;
```

Target PDCCH IBLER optimization

```
//Configuring the target PDCCH IBLER  
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, PdcchBlerTarget=1;
```

PDCCH CCE optimization

```
//Setting the NRDUCellPdcch.OccupiedSymbolNum parameter to 2SYM (Modification of this parameter will  
cause the cell to automatically restart.)  
MOD NRDUCELLPDCCH: NrDuCellId=1, OccupiedSymbolNum=2SYM;  
//Turning on the switch for PDCCH uplink-to-downlink CCE ratio adaptation  
MOD NRDUCELLPDCCH: NrDuCellId=1, PdcchAlgoSwitch=UL_DL_CCE_RATIO_ADAPT_SW-1;  
//Turning on the switch for SR-based smart scheduling  
MOD NRDUCELLPUSCH: NrDuCellId=1, UIPuschAlgoSwitch=SR_SMART_SCH_SW-1;
```

PDCCH aggregation level adjustment mechanism optimization

```
//Configuring the minimum PDCCH aggregation level  
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, PdcchMinAggLvl=AGGLVL4;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

PDCCH CCE optimization

```
//Turning off the switch for PDCCH uplink-to-downlink CCE ratio adaptation  
MOD NRDUCELLPDCCH: NrDuCellId=1, PdcchAlgoSwitch=UL_DL_CCE_RATIO_ADAPT_SW-0;  
//Turning off the switch for SR-based smart scheduling  
MOD NRDUCELLPUSCH: NrDuCellId=1, UIPuschAlgoSwitch=SR_SMART_SCH_SW-0;
```

PDCCH aggregation level adjustment mechanism optimization

```
//Setting PdcchMinAggLvl to NOT_CONFIG  
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, PdcchMinAggLvl=NOT_CONFIG;
```

Switch for basic high-reliability functions

```
//Turning off the switch for basic high-reliability functions  
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, HighReliabilitySwitch=HIGH_RELIABILITY_BASIC_SW-0;
```

6.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.2.4.2 Activation Verification

To check whether functions related to control channel reliability optimization have been activated, perform real-time UE-level performance monitoring as follows:

- Step 1** After a UE accesses a cell, enable the UE to set up a high-reliability-oriented bearer. Move the UE to the cell center and enable the UE to perform a downlink ping service instead of an uplink service.
- Step 2** On the MAE-Access, start UE-level channel quality monitoring (corresponding to **Quality of Channel Monitoring** on the GUI) and MCS-based count monitoring (corresponding to **MCS Count Monitoring** on the GUI).
- Step 3** Observe the UE channel quality by **Quality of Channel Monitoring** and the number of uplink scheduling times by **MCS Count Monitoring**. If the number of

uplink scheduling times increases, functions related to control channel reliability optimization have taken effect.

----End

6.2.4.3 Network Monitoring

For details, see [4.4.3 Network Monitoring](#).

7 Functions Oriented Toward Increasing the Number of Supported UEs Requiring Connectivity with Low Latency and High Reliability

A network can support a greater number of UEs requiring connectivity with low latency and high reliability by reducing resource consumption or improving resource utilization.

7.1 PLC UE Number Specification Improvement

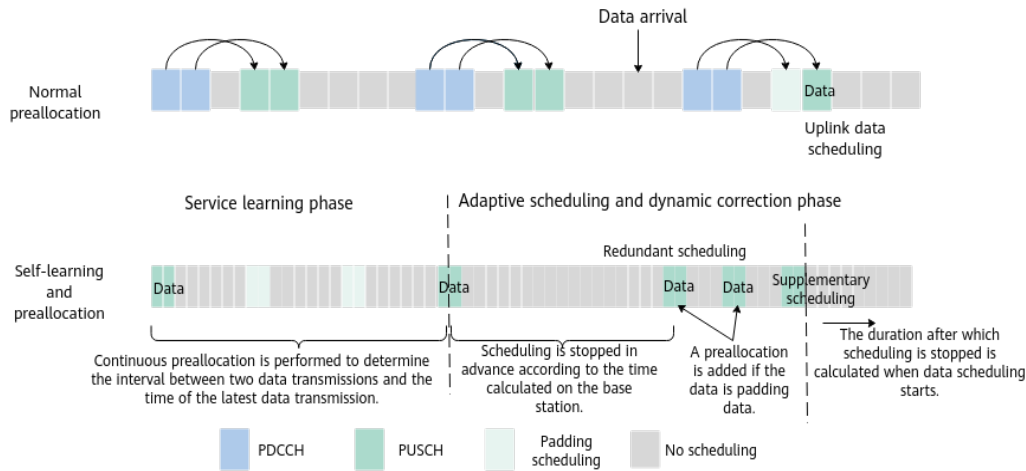
7.1.1 Principles

To increase the number of online PLC UEs supported on a network, the PLC UE number specification improvement function is introduced. [Figure 7-1](#) illustrates how this function works. After this function is enabled, the following mechanisms apply:

- Self-learning-based preallocation enables the gNodeB to use a large amount of data for training. As a result of this training, the gNodeB adaptively learns the time at which a UE transmits a service packet, calculates an interval between packet transmissions, and performs intermittent preallocation according to the interval. This type of preallocation reduces uplink resource consumption, thereby increasing the number of online PLC UEs supported on a network.
- In the intermittent preallocation phase, the gNodeB ensures delay and reliability performance in the event of packet transmission jitter. This is achieved through redundant and supplementary scheduling, as well as the introduction of an additional scheduling opportunity following the cessation of scheduling.

This function is controlled by the **UL_PREALLOC_SELFSTUDY_QCI_SW** option of the **gNBDUSchParamGroup.ScheduleOptSwitch** parameter.

Figure 7-1 Mechanism of PLC UE number specification improvement



7.1.2 Network Analysis

7.1.2.1 Benefits

PLC UE number specification improvement increases the uplink capacity of services requiring connectivity with high reliability by 40% and reduces the resource consumption of control and data channels by about 15%.

Conditions for Optimal Gains

For details, see [Conditions for Optimal Gains](#).

7.1.2.2 Impacts

The impacts between this function and other functions are described in [7.1.3.2.3 Function Impacts](#).

There are no impacts on the network.

7.1.3 Requirements

7.1.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-051206	URLLC Introduction	NR0SURLLCP00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

7.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.1.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Service-differentiated low latency	SERV_DIFF_LOW_LATENCY_SW option of the NRDUCellAlgorithmSwitch.LowLatencySwitch parameter	<i>Service-differentiated Experience Guarantee</i>	PLC UE number specification improvement depends on service-differentiated low latency.

7.1.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	QCI-based uplink smart preallocation	UL_SMART_PREALLOC_QCI_SW option of the gNBDSchParamGroup.ScheduleOptSwitch parameter	<i>Service-differentiated Experience Guarantee</i>	These two functions cannot be both enabled: PLC UE number specification improvement (controlled by the UL_PREALLOC_SELFSTUDY_QCI_SW option of the gNBDSchParamGroup.ScheduleOptSwitch parameter) and QCI-based uplink smart preallocation.

7.1.3.2.3 Function Impacts

None

7.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

- Main control boards: Supported models are limited to the UMPTe/UMPTg/UMPTHa/UMPTHb series.
- Baseband processing units: Supported models are limited to the UBBPg and later series (NR-only mode).

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

7.1.3.4 Cells

For details, see [4.3.4 Cells](#). Additionally, PLC UE number specification improvement is not supported when **NRDUCell.SlotAssignment** is set to **2_3_DSUUU**.

7.1.3.5 Others

None

7.1.4 Operation and Maintenance

7.1.4.1 Data Configuration

7.1.4.1.1 Data Preparation

[Table 7-1](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 7-1 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Schedule Option Switch	gNBDUSchParamGroup.ScheduleOptSwitch	UL_PREALLOC_SELFSTUDY_QCI_SW	To enable PLC UE number specification improvement, select this option.

7.1.4.1.2 Using MML Commands

Before using MML commands, refer to [7.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling PLC UE number specification improvement
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1,
ScheduleOptSwitch=UL_PREALLOC_SELFSTUDY_QCI_SW-1;
//(Optional) Configuring the interval and data volume for uplink preallocation
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, UIPreallocationInterval=32,
UIPreallocationDataVolume=600;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling PLC UE number specification improvement
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1,
ScheduleOptSwitch=UL_PREALLOC_SELFSTUDY_QCI_SW-0;
```

7.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.1.4.2 Activation Verification

Run the **LST GNBDUSCHPARAMGROUP** command to query the **QCI-based Uplink Precise Preallocate Sw** setting of the **Schedule Option Switch** parameter to see whether the corresponding function has been enabled.

7.1.4.3 Network Monitoring

For details, see [4.4.3 Network Monitoring](#).

7.2 Scheduling Based on Delay-Sensitive Requirements

7.2.1 Principles

In a cell that serves multiple UEs requiring connectivity with high reliability, limited uplink resources make it impossible to guarantee delay for all these UEs. In addition, delay requirements vary depending on the specific type of UE. To meet the delay requirements of a larger number of UEs when resources are limited, scheduling based on delay-sensitive requirements is introduced.

This function allows configuration of a data transmission period for the URLLC service running on a UE based on the UE's service model. This enables the provision of differentiated delay guarantee for UEs requiring connectivity with high reliability. A shorter data transmission period for the URLLC service means a higher delay requirement.

To meet the specific delay requirements of UEs in a high-reliability-oriented scheduling queue, the UEs' scheduling priorities are adjusted. The scheduling priority of a UE requiring connectivity with high reliability is related to two factors: URLLC data transmission period and waiting time before scheduling. To illustrate, a shorter URLLC data transmission period and a longer waiting time before scheduling indicate a higher scheduling priority. A shorter URLLC data transmission period leads to a higher scheduling priority, based on the assumption that the waiting time before scheduling is the same. This is illustrated in [Figure 7-2](#).

The following UEs can be added to a high-reliability-oriented scheduling queue:

- UEs requiring connectivity with low latency and high reliability
- Delay-sensitive UEs that do not require connectivity with high reliability and are not scheduled for a long time

The **gNBDUSchParamGroup.CycleTime** parameter specifies the URLLC data transmission period. This parameter can be set to a value other than **NOT_CONFIG** to enable scheduling based on delay-sensitive requirements.

Figure 7-2 Scheduling based on delay-sensitive requirements (based on the assumption that the waiting time before scheduling is the same)



 NOTE

If a QCI is not defined in the **NRDUCellQciBearer** MO, the **gNBDUSchParamGroup.CycleTime** parameter for the QCI uses the value **NOT_CONFIG**.

7.2.2 Network Analysis

7.2.2.1 Benefits

Scheduling based on delay-sensitive requirements satisfies the delay and reliability requirements of multiple UEs in scenarios where services with different delay requirements are performed.

Conditions for Optimal Gains

For details, see [Conditions for Optimal Gains](#).

7.2.2.2 Impacts

The impacts between this function and other functions are described in [7.2.3.2.3 Function Impacts](#).

There are no impacts on the network.

7.2.3 Requirements

7.2.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-0512 06	URLLC Introduction	NR0SURLLCP00	per Cell

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

7.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.2.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Identification of UEs requiring connectivity with low latency and high reliability	HIGH_RELIABILITY_BASIC_SWITCH option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter gNBDUMacParamGroup.SchedulingType set to HIGH_RELIABILITY_SCH gNBQciBearer.ServiceType set to URLLC_HIGH_RELIABILITY_SERVICE	<i>URLLC</i>	Scheduling based on delay-sensitive requirements depends on identification of UEs requiring connectivity with low latency and high reliability.
Low-frequency TDD	Rank configuration for low latency and high reliability	gNBDUSchParamGroup.UllcMaxRank	<i>URLLC</i>	If the gNBDUSchParamGroup.CycleTime parameter is set to MS8 , the gNBDUSchParamGroup.UllcMaxRank parameter can only be set to RANK1 .

7.2.3.2.2 Mutually Exclusive Functions

None

7.2.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink PDB-based differentiated scheduling	gNBQciBeare r.PdbDiffSchP olicy set to a value other than OFF	None	When both scheduling based on delay-sensitive requirements and downlink PDB-based differentiated scheduling are enabled, only downlink PDB-based differentiated scheduling takes effect.

7.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

- Main control boards: Supported models are limited to the UMPTe/UMPTg/UMPTa/UMPTb series.
- Baseband processing units: Supported models are limited to the UBBPg and later series (NR-only mode).

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

7.2.3.4 Cells

For details, see [4.3.4 Cells](#).

7.2.3.5 Others

None

7.2.4 Operation and Maintenance

7.2.4.1 Data Configuration

7.2.4.1.1 Data Preparation

Table 7-2 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 7-2 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Cycle Time	gNBDUSchParamGroup.CycleTime	None	Set this parameter based on the network plan.

7.2.4.1.2 Using MML Commands

Before using MML commands, refer to **7.2.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Adding a gNodeB DU scheduling parameter group
ADD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1;
//Configuring the period
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, CycleTime=MS20;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Leaving the period not configured
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, CycleTime=NOT_CONFIG;
```

7.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.2.4.2 Activation Verification

Run the **LST GNBUSCHPARAMGROUP** command. If the value of **Cycle Time(ms)** is not **NOT_CONFIG** in the command output, the corresponding function has been enabled.

7.2.4.3 Network Monitoring

For details, see [4.4.3 Network Monitoring](#).

8 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

9 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

10 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

11 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.214: "NR; Physical layer procedures for data"
- 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"
- *Carrier Aggregation*
- *Modulation Schemes*
- *High Speed Mobility*
- *UL and DL Decoupling*
- *MIMO (TDD)*
- *Super Uplink*
- *Hyper Cell*
- *VoNR*
- *Uplink Scheduling*
- *Downlink Scheduling*
- *UE Power Saving*
- *Network Slicing*
- *Multi-Operator Sharing*
- *FWA*
- *DRX*
- *Energy Conservation and Emission Reduction*
- *Interference Avoidance*
- *EPC-based NSA Basics*
- *5G Networking and Signaling*
- *Channel Management*
- *Coordinated Interference Management (Low-Frequency TDD)*
- *SA Mobility Management in Connected Mode*
- *NSA Mobility Management*
- *Mobility Load Balancing in Connected Mode*
- *Multi-RAT Coordinated Energy Saving*

- *Admission Control*
- *3D Networking Experience Improvement*
- *Green Site*
- *Massive MIMO iBeam (Low-Frequency TDD)*
- *Massive MIMO Uplink Boosting (Low-Frequency TDD)*
- *Turbo Uplink (Low-Frequency TDD)*
- *Power Control*
- *RedCap*
- *Resource Control in NSA Networking*
- *License Control Item Lists*
- *Multi-Frequency Smart Aggregation*
- *Stable Video Surveillance Experience*
- *URLLC for Vehicle Communications*
- *QCI-based Resource Control*
- *Service-differentiated Experience Guarantee*